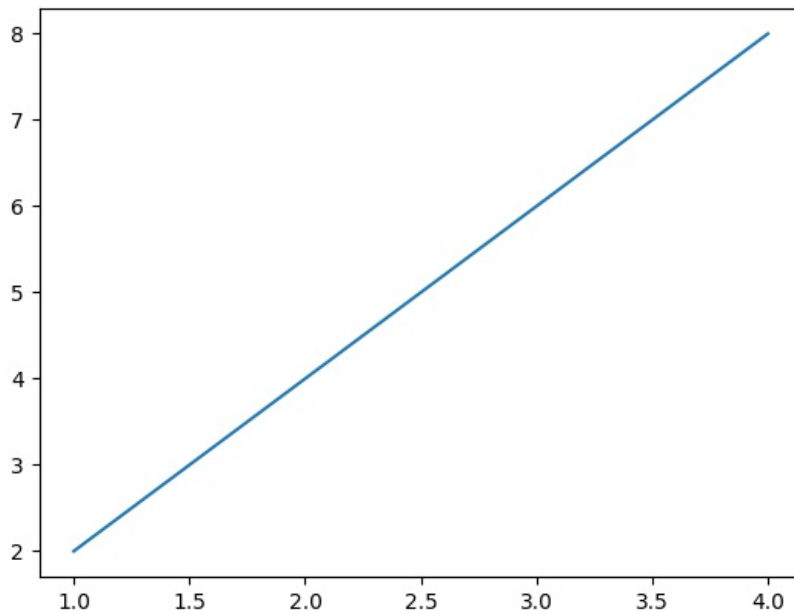


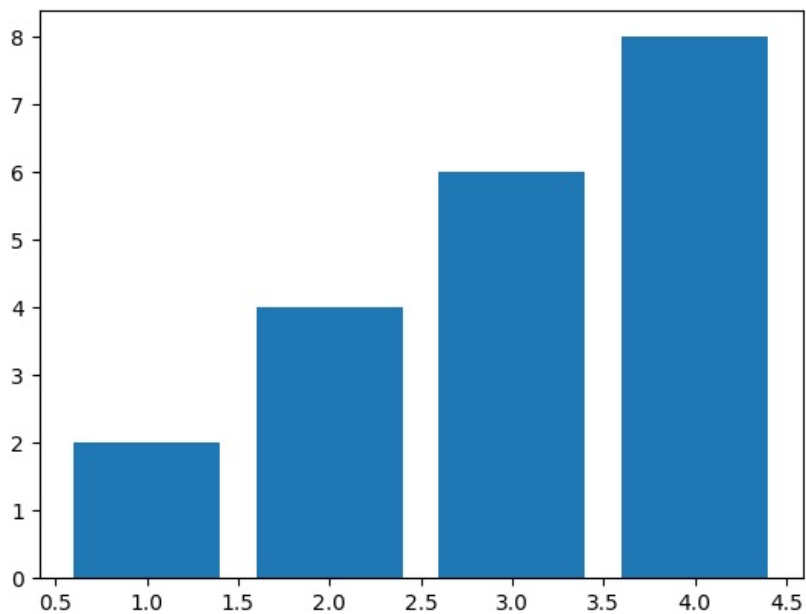
Linear Plot

```
In [27]: import matplotlib.pyplot as plt
x=[1,2,3,4]
y=[2,4,6,8]
plt.plot(x,y)
plt.show()
```



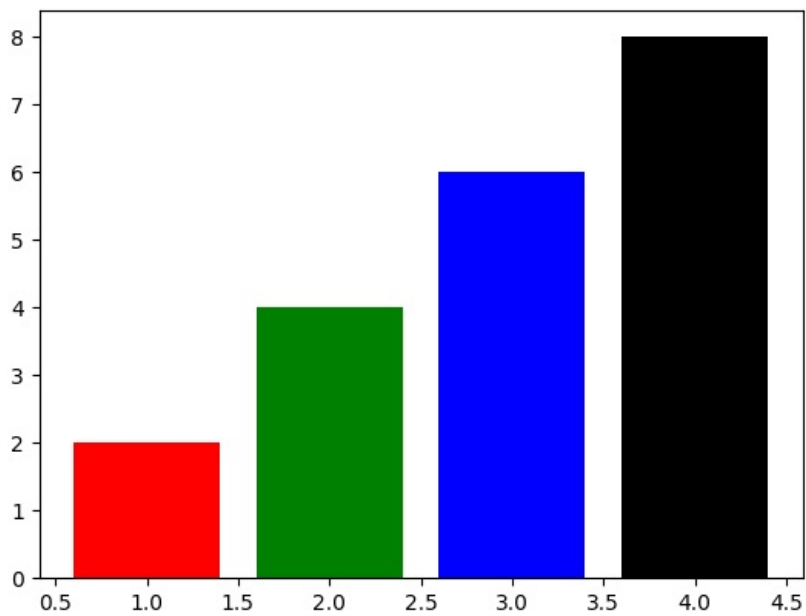
Bar Graph

```
In [28]: plt.bar(x,y)
plt.show()
```



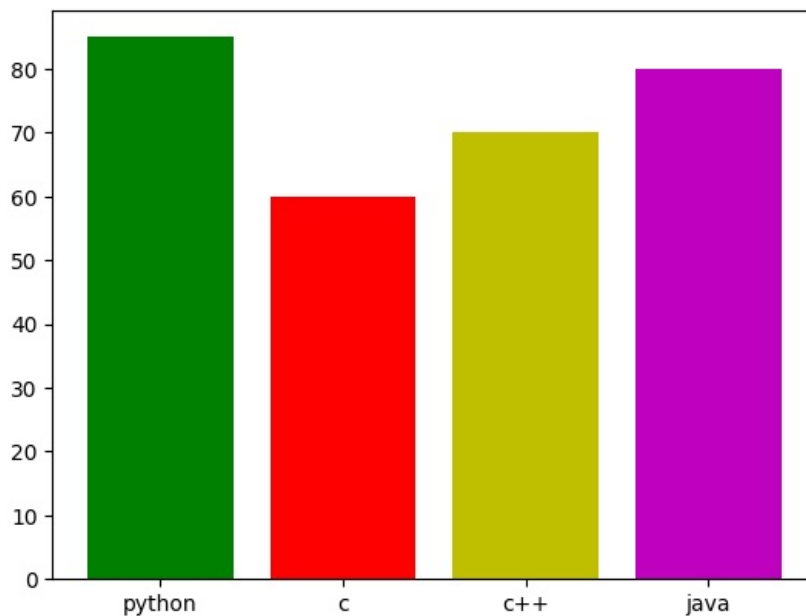
Color

```
In [29]: c=["r","g","b","k"]
plt.bar(x,y,color=c)
plt.show()
```



Bar Plot

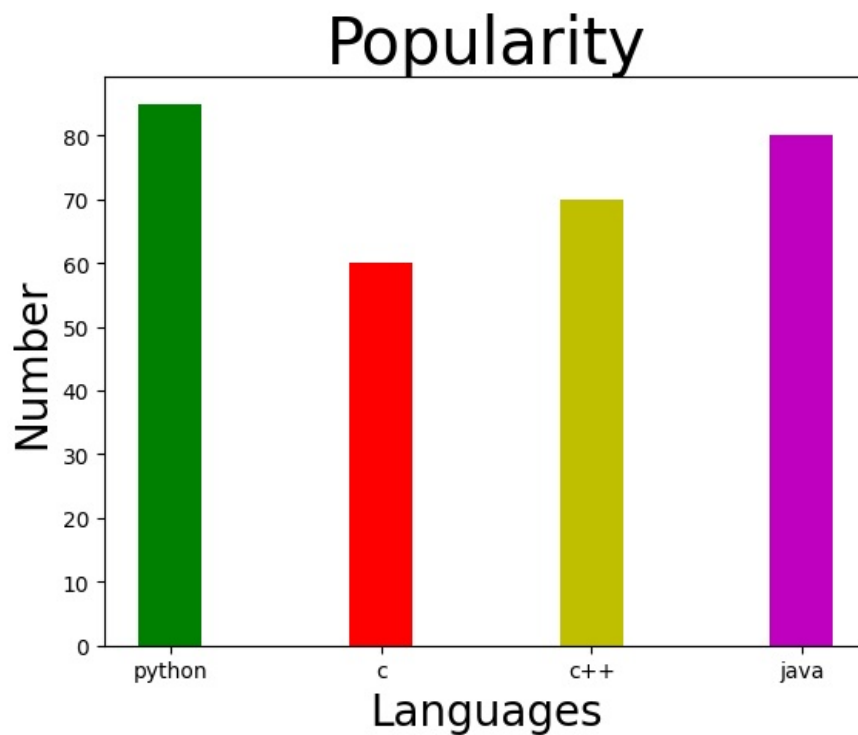
```
In [30]: import matplotlib.pyplot as plt
x=["python","c","c++","java"]
y=[85,60,70,80]
c=["g","r","y","m"]
plt.bar(x,y,color=c)
plt.show()
```



Label

```
In [31]: plt.xlabel("Languages",fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity",fontsize=30)
plt.bar(x,y,color=c, width=0.3)
```

```
Out[31]: <BarContainer object of 4 artists>
```



plt.yticks(range(0, 91, 5))

91 exclusive hai

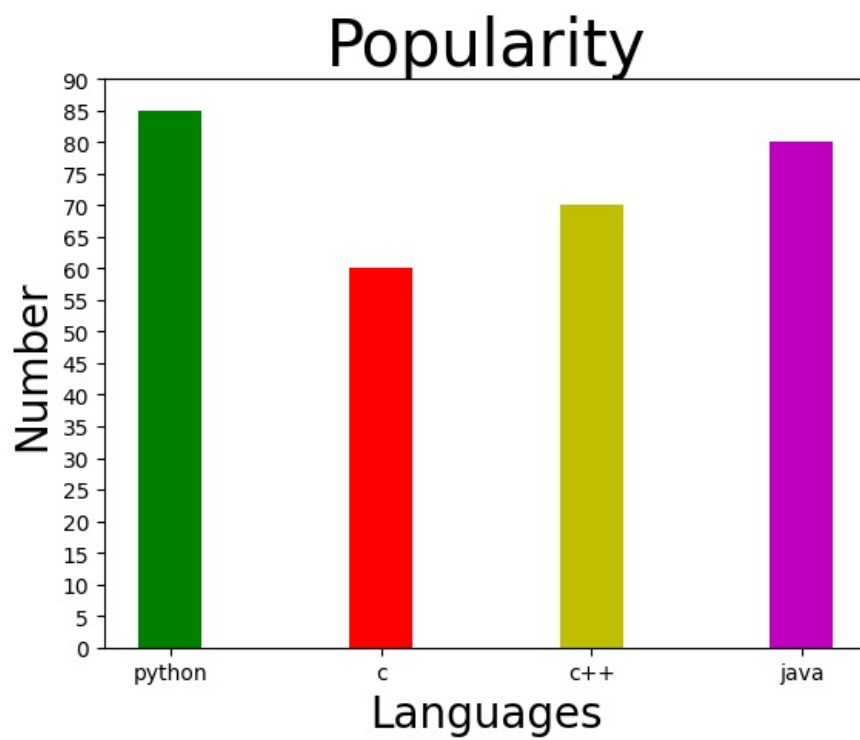
5 ka Difference in Y-axis me

```
In [32]: import matplotlib.pyplot as plt
x=["python","c","c++","java"]
y=[85,60,70,80]
c=["g","r","y","m"]
plt.xlabel("Languages",fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity",fontsize=30)
plt.bar(x,y,color=c, width=0.3)

plt.yticks(range(0, 91, 5))

# 91 exclusive hai,

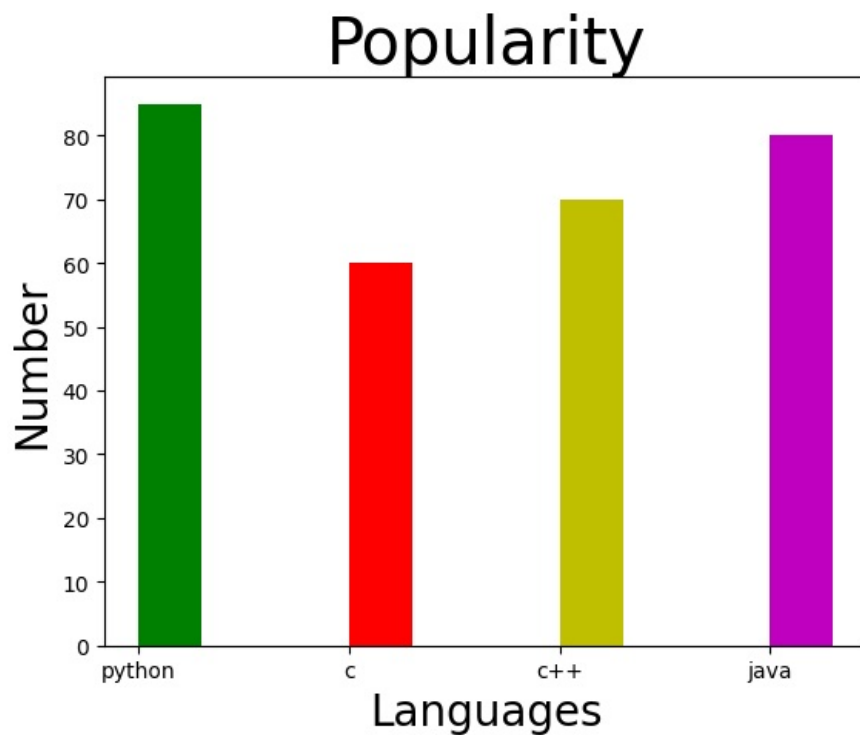
plt.show()
```



Edge

```
In [33]: plt.xlabel("Languages", fontsize=20 )  
plt.ylabel("Number", fontsize=20)  
plt.title("Popularity", fontsize=30)  
plt.bar(x,y,color=c, width=0.3,align="edge")
```

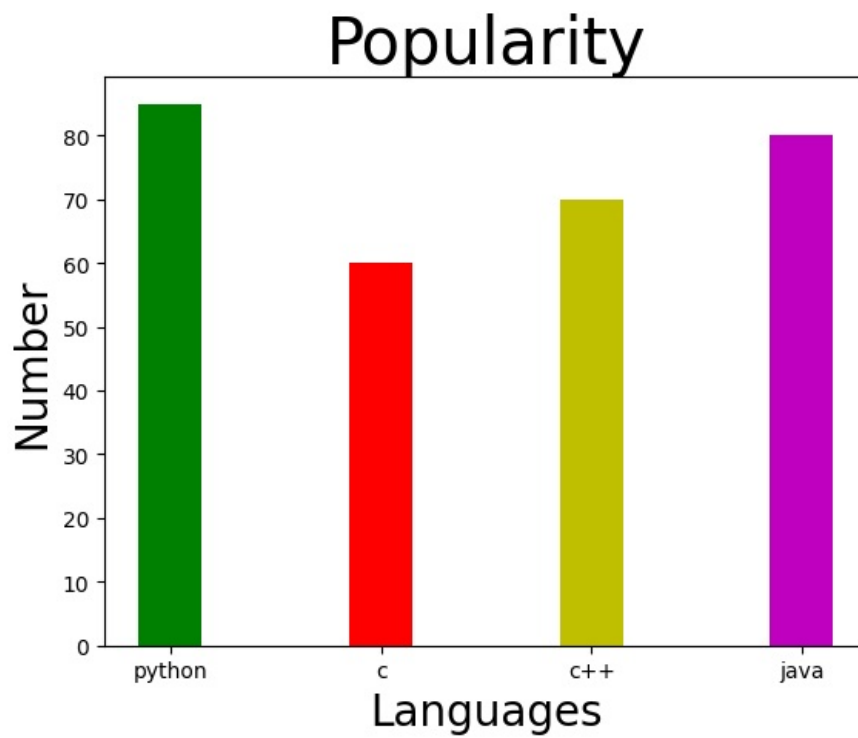
Out[33]: <BarContainer object of 4 artists>



Center

```
In [34]: plt.xlabel("Languages", fontsize=20 )  
plt.ylabel("Number", fontsize=20)  
plt.title("Popularity", fontsize=30)  
  
plt.bar(x,y,color=c, width=0.3,align="center")
```

Out[34]: <BarContainer object of 4 artists>



Edge Color Black

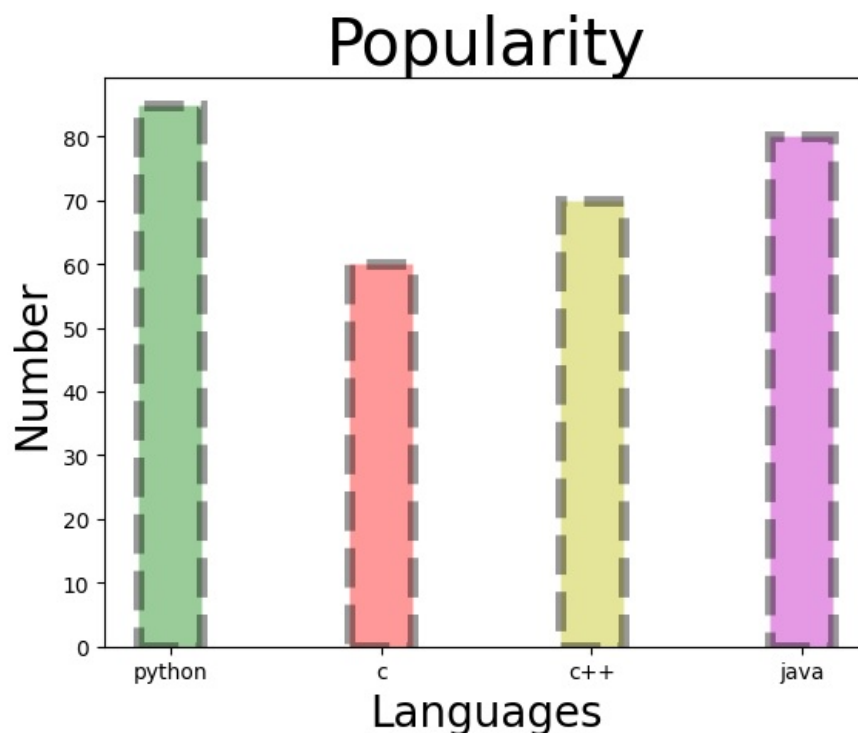
Line width

Line style

alpha parameter to make graph light color

```
In [35]: plt.xlabel("Languages", fontsize=20 )  
plt.ylabel("Number", fontsize=20)  
plt.title("Popularity", fontsize=30)  
  
plt.bar(x,y,color=c, width=0.3, align="center", edgecolor="k", linewidth=5, linestyle="dashed", alpha=0.4)
```

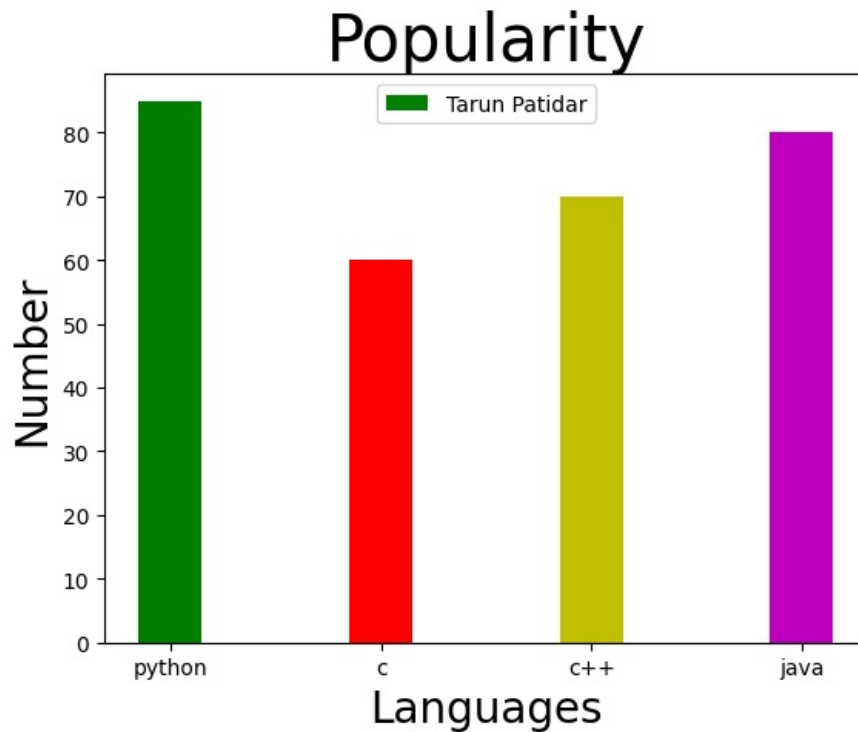
Out[35]: <BarContainer object of 4 artists>



Legend

```
In [36]: plt.xlabel("Languages", fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity", fontsize=30)

plt.bar(x,y,color=c, width=0.3,align="center", label="Tarun Patidar")
plt.legend(loc="upper center")
plt.show()
```



Multiple Bar Graph

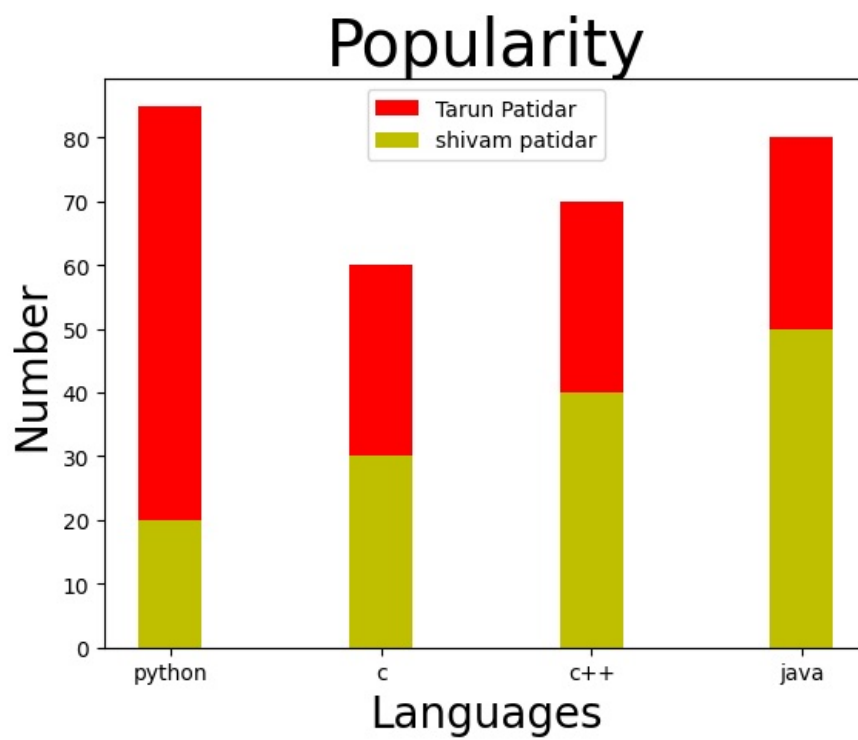
```
In [2]: import matplotlib.pyplot as plt
x=["python","c","c++","java"]
y=[85,60,70,80]
z=[20,30,40,50]

plt.xlabel("Languages", fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity", fontsize=30)

plt.bar(x,y,color="r", width=0.3,align="center", label="Tarun Patidar")

plt.bar(x,z,color="y", width=0.3,align="center", label="shivam patidar")

plt.legend(loc="upper center")
plt.show()
```



Multiple Bar Graph

```
In [10]: import matplotlib.pyplot as plt
import numpy as np
x=["python","c","c++","java"]
y=[85,60,70,80]

z=[20,30,40,50]

plt.xlabel("Languages",fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity",fontsize=30)

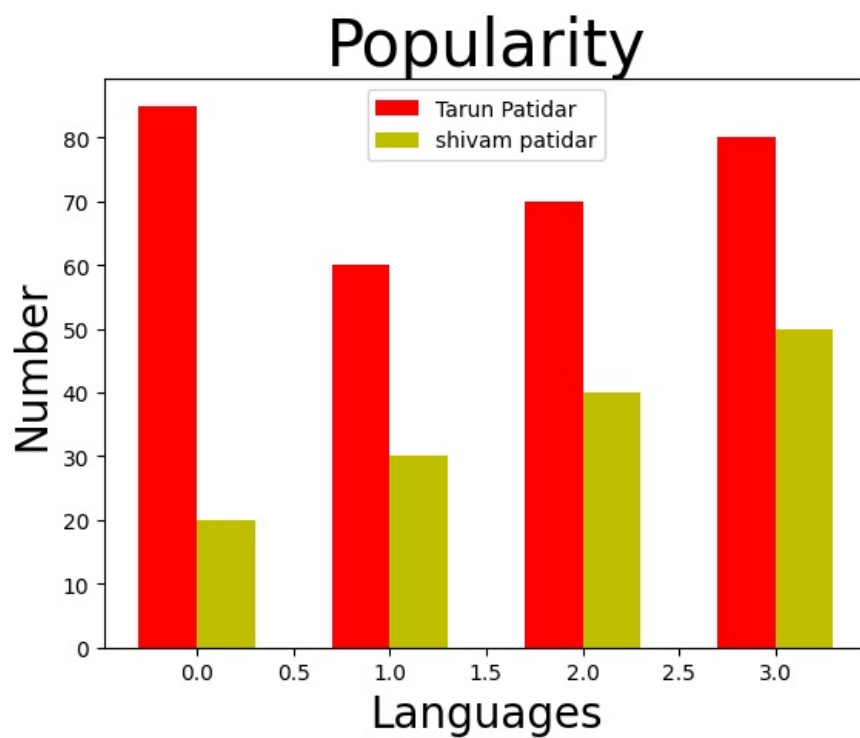
p=np.arange(len(x))

width=0.3

plt.bar(p-width/2,y,color="r",width=width,align="center", label="Tarun Patidar")
#left    p-width/2

plt.bar(p+width/2,z,color="y",width=width,align="center", label="shivam patidar")
#right   p+width/2

plt.legend(loc="upper center")
plt.show()
```



Multiple Bar Graph

`plt.xticks(p+width/2,x)`

```
In [14]: import matplotlib.pyplot as plt
import numpy as np
x=["python","c","c++","java"]
y=[85,60,70,80]

z=[20,30,40,50]

plt.xlabel("Languages",fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity",fontsize=30)

p=np.arange(len(x))

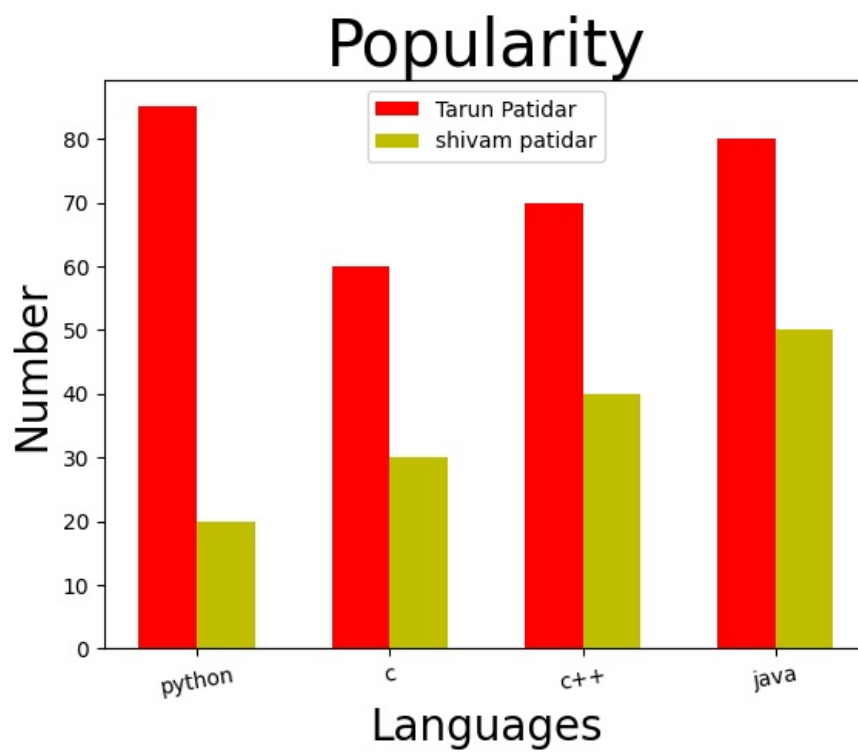
p1=[j+width for j in p]
width=0.3

plt.bar(p,y,color="r",width=width,align="center", label="Tarun Patidar")

plt.bar(p1,z,color="y",width=width,align="center", label="shivam patidar")
#right    p1=[j+width for j in p]

plt.legend(loc="upper center")
#location parameter

plt.xticks(p+width/2,x,rotation=10)
plt.show()
```

Label

```
In [25]: import matplotlib.pyplot as plt
import numpy as np
x=["python","c","c++","java"]
y=[85,60,70,80]
z=[20,30,40,50]

plt.xlabel("Languages",fontsize=20 )
plt.ylabel("Number", fontsize=20)
plt.title("Popularity",fontsize=30)

p=np.arange(len(x))
width=0.3
p1=[j+width for j in p]

plt.barh(p,y,color="r",height=width,align="center", label="Tarun Patidar")

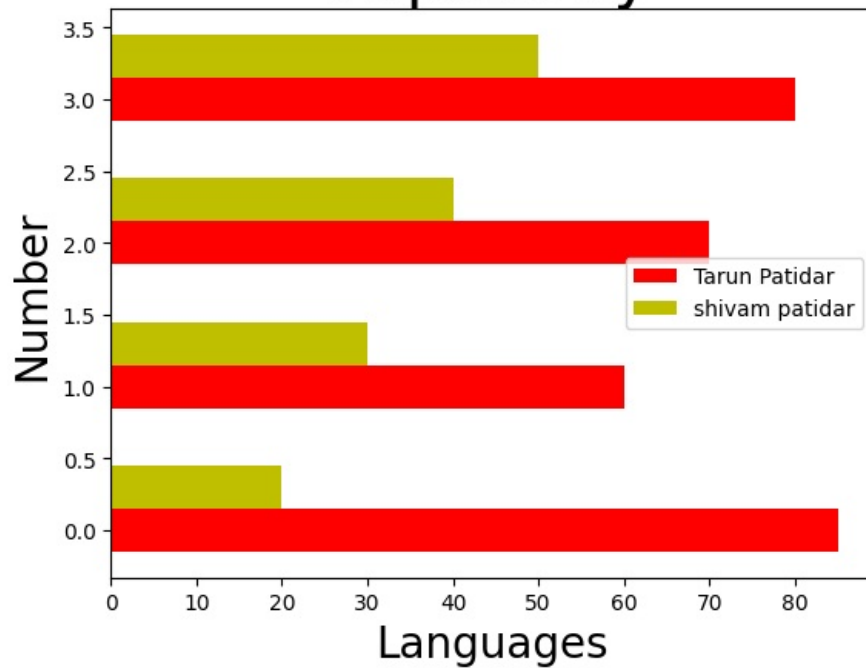
plt.barh(p1,z,color="y",height=width,align="center", label="shivam patidar")
#right p1=[j+width for j in p]

plt.legend(loc="center right", ncol=1 )

# chart ke andar right side center mein
# top to bottom

plt.show()
```

Popularity



Scatter Plot

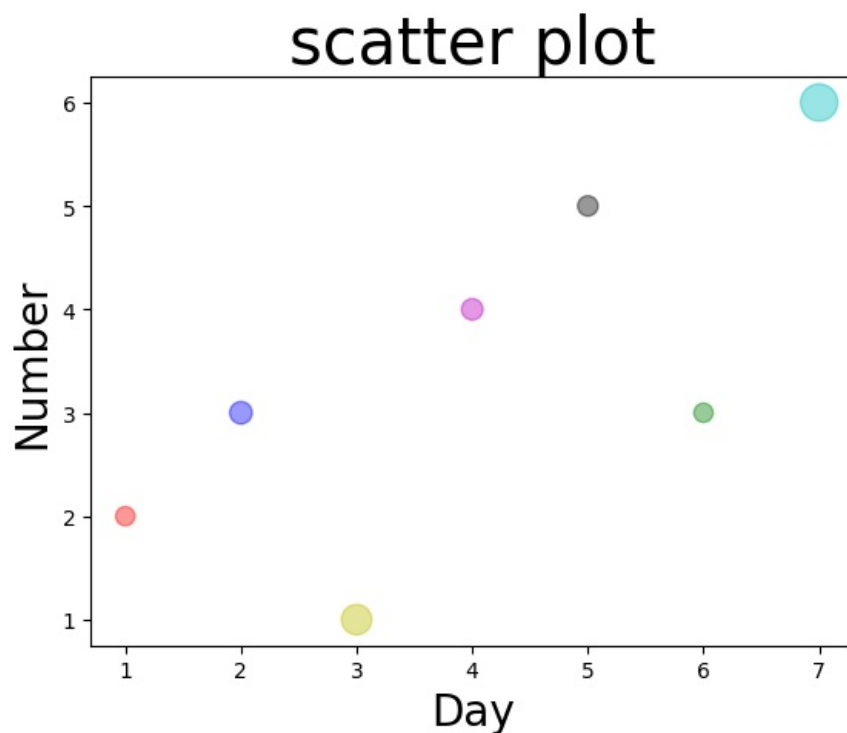
```
In [30]: import matplotlib.pyplot as plt

day=[1,2,3,4,5,6,7]
num=[2,3,1,4,5,3,6]

c=["r","b","y","m","k","g","c"]
sizes=[80,110,200,100,90,80,300]

plt.title("scatter plot",fontsize=30)
plt.xlabel("Day",fontsize=20)
plt.ylabel("Number",fontsize=20)

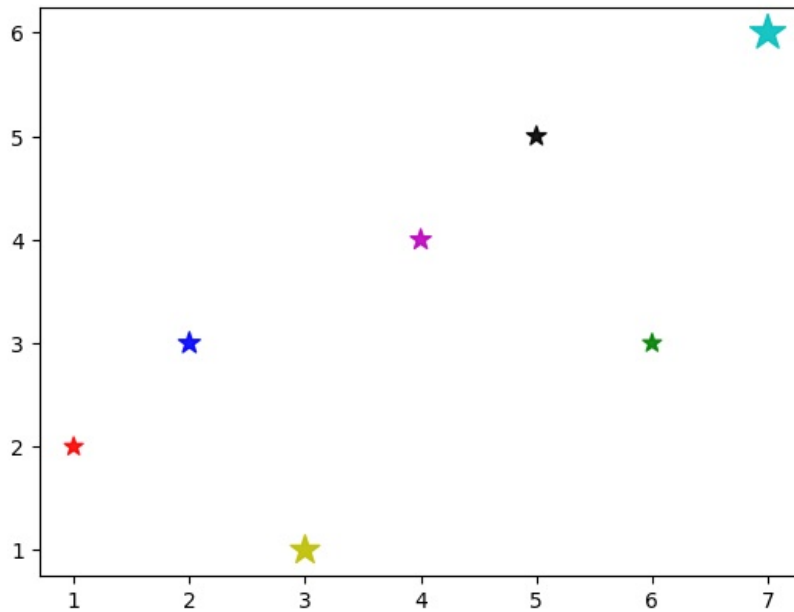
plt.scatter(day,num,color=c,s=sizes,alpha=0.4)
plt.show()
```



Marker parmeter

```
In [32]: plt.scatter(day,num,color=c,s=sizes,alpha=0.9,marker="*")
```

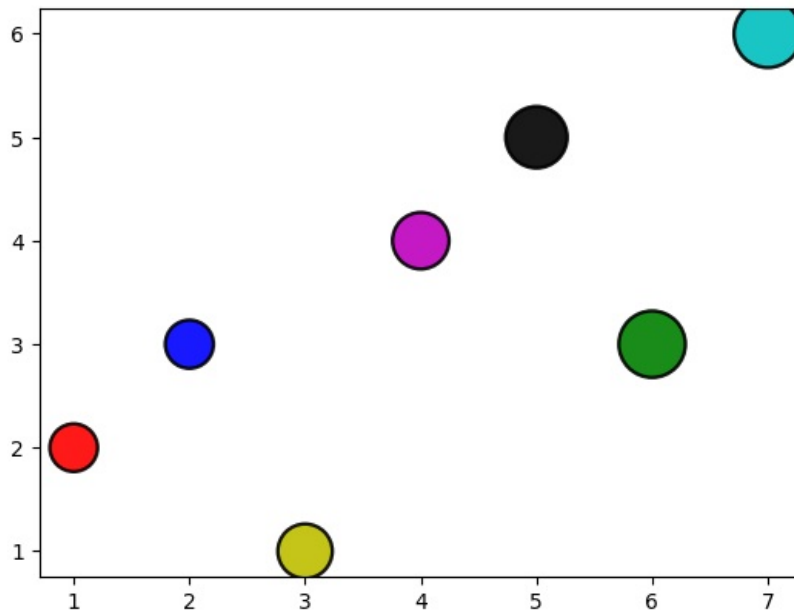
```
Out[32]: <matplotlib.collections.PathCollection at 0x1d518fc3750>
```



edgecolor, linewidth

```
In [42]: sizes=[500,510,650,700,830,970,1000]  
plt.scatter(day,num,color=c,s=sizes,alpha=0.9,edgecolor="k",linewidth=1.7)
```

```
Out[42]: <matplotlib.collections.PathCollection at 0x1d5198bac10>
```



plt.colorbar()

```
In [51]: import matplotlib.pyplot as plt  
  
day=[1,2,3,4,5,6,7]  
num=[2,3,1,4,5,3,6]  
  
colors=[10,49,30,29,50,20,30]  
  
# color 0 to 100  
  
sizes=[300,410,500,600,900,880,600]  
  
plt.title("scatter plot",fontsize=30)  
plt.xlabel("Day",fontsize=20)  
plt.ylabel("Number",fontsize=20)
```

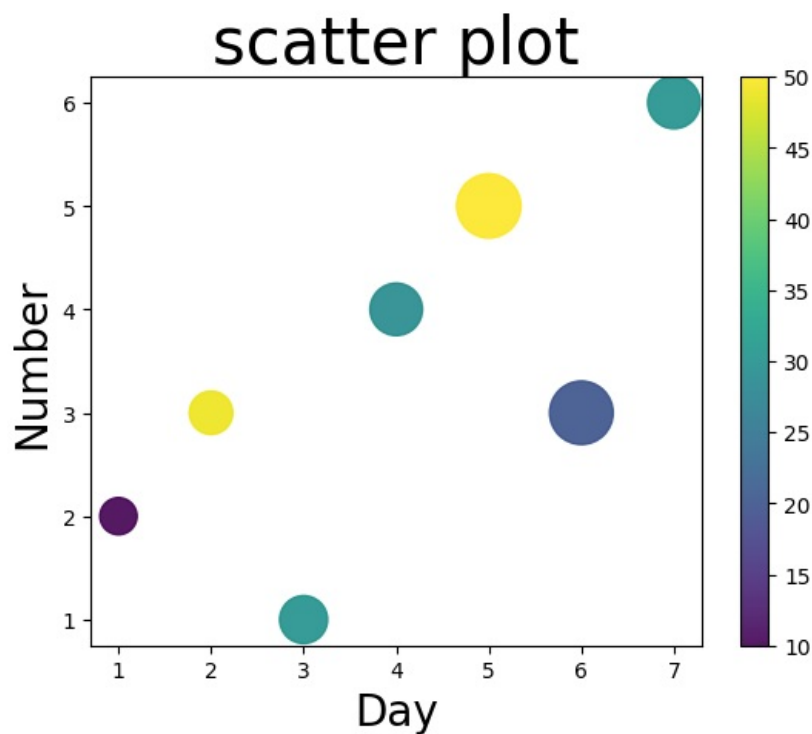
```
plt.scatter(day,num,c=colors,s=sizes,alpha=0.9,cmap="viridis")

#plt.scatter(day,num,c=colors,s=sizes,alpha=0.9,cmap="Accent")

#cmap => color map parameter

plt.colorbar()

plt.show()
```



plt.colorbar()

In [53]: `import matplotlib.pyplot as plt`

```
day=[1,2,3,4,5,6,7]
num=[2,3,1,4,5,3,6]

colors=[10,49,30,29,50,20,30]

# color 0 to 100

sizes=[300,410,500,600,900,880,600]

plt.title("scatter plot",fontsize=30)
plt.xlabel("Day",fontsize=20)
plt.ylabel("Number",fontsize=20)

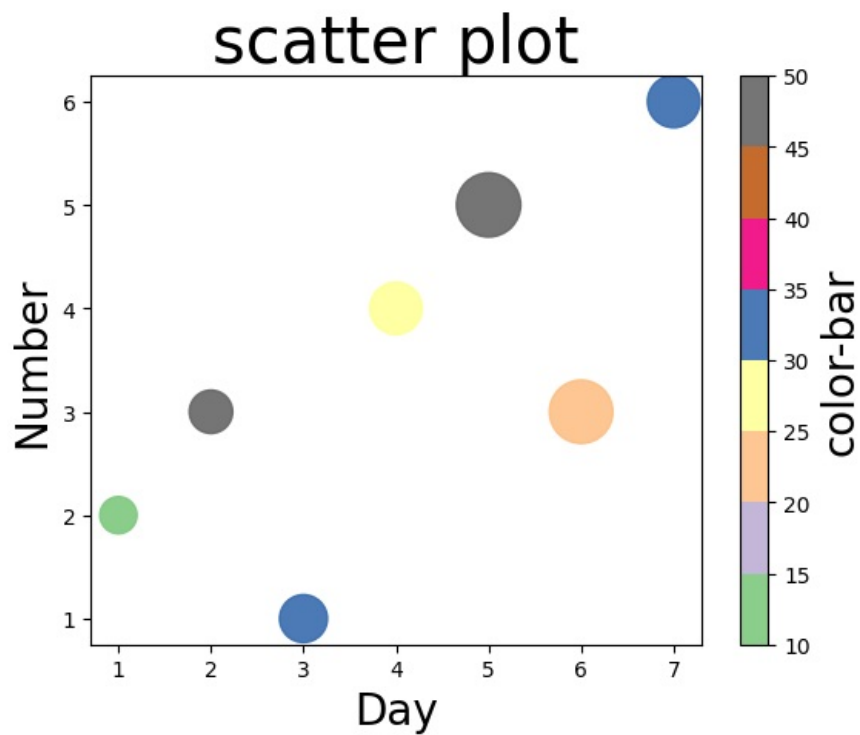
#plt.scatter(day,num,c=colors,s=sizes,alpha=0.9,cmap="viridis")

plt.scatter(day,num,c=colors,s=sizes,alpha=0.9,cmap="Accent")

#cmap => color map parameter

t= plt.colorbar()
t.set_label("color-bar",fontsize=20)

plt.show()
```



Multiple scatter dots

```
In [54]: import matplotlib.pyplot as plt

day=[1,2,3,4,5,6,7]
num=[2,3,1,4,5,3,6]

num2=[3,2,4,5,6,2,3]

colors=[10,49,30,29,50,20,30]

# color 0 to 100

sizes=[300,410,500,600,900,880,600]

plt.title("scatter plot",fontsize=30)
plt.xlabel("Day",fontsize=20)
plt.ylabel("Number",fontsize=20)

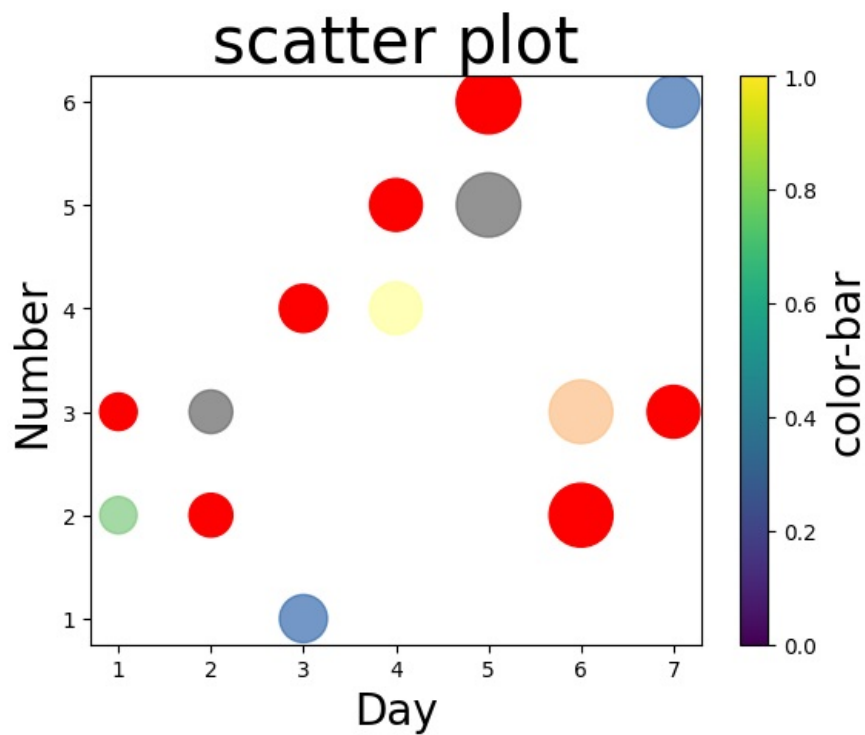
#plt.scatter(day,num,c=colors,s=sizes,alpha=0.9,cmap="viridis")

plt.scatter(day,num,c=colors,s=sizes,alpha=0.7,cmap="Accent")
plt.scatter(day,num2,color="r",s=sizes,alpha=1)

#cmap => color map parameter

t= plt.colorbar()
t.set_label("color-bar",fontsize=20)

plt.show()
```



Histogram

```
In [10]: import matplotlib.pyplot as plt
import numpy as np

import random
x=np.random.randint(10,60,(50))

print(x)
```

```
[38 47 41 26 25 48 54 28 14 36 42 40 32 22 49 39 17 27 10 26 57 34 10 41
 29 22 26 50 36 19 37 55 10 37 45 53 42 19 20 20 11 29 27 45 32 47 23 23
 25 28]
```

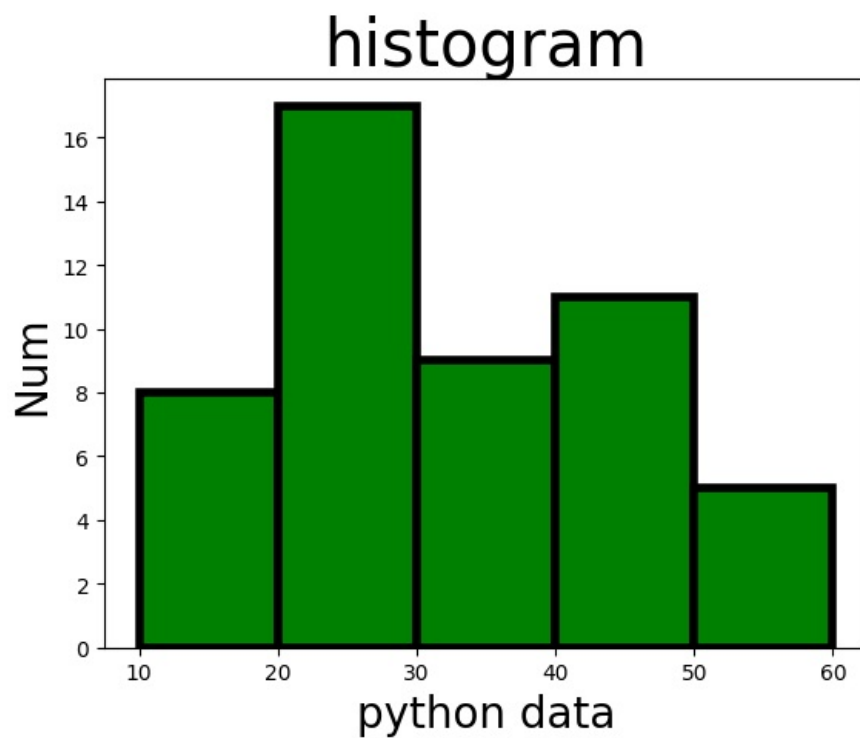
Histogram

```
In [21]: num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
 29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
 25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

bins=[10,20,30,40,50,60]
plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=4,linestyle="solid",bins=bins )
plt.show()
```



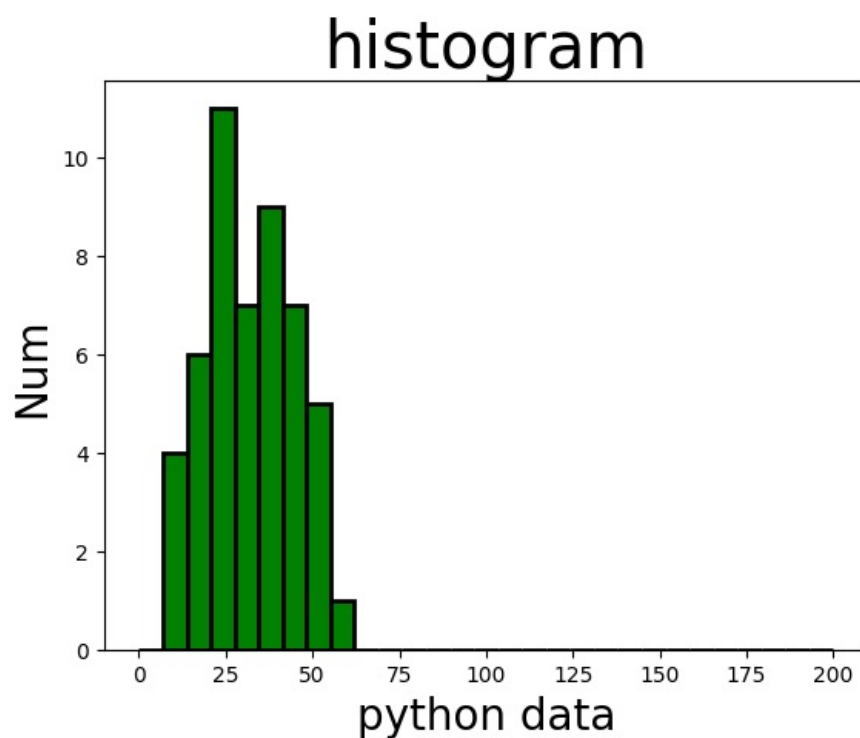
bins="auto", range=(0,200)

```
In [28]: num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bins="auto", range=(0,200) )
plt.show()
```



Reverse the Frequency by comulative parameter

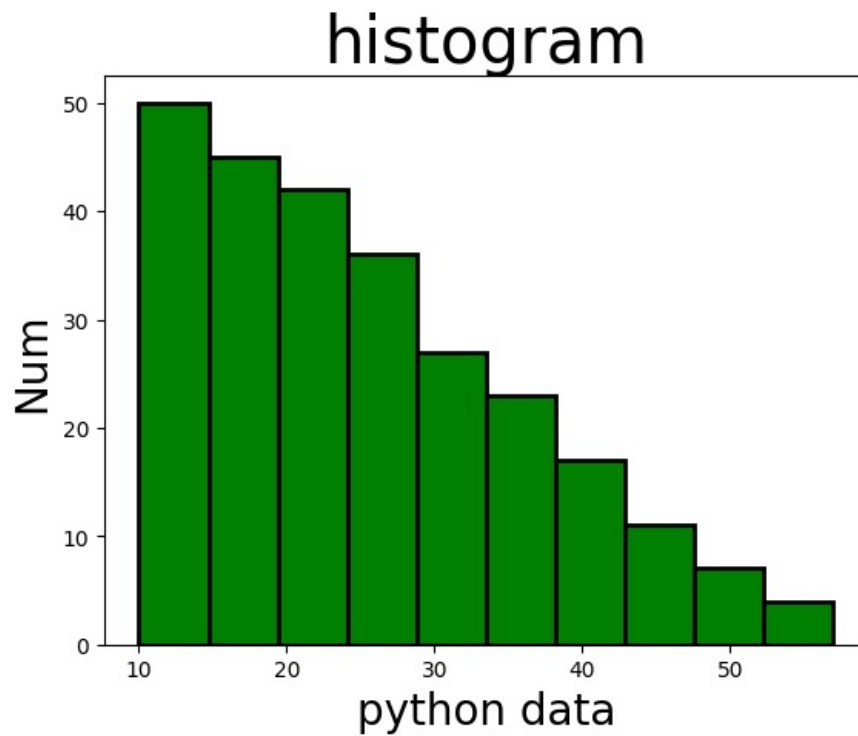
```
In [34]: num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
```

```
25, 28]
```

```
plt.title("histogram", fontsize=30)
plt.xlabel("python data", fontsize=20)

plt.ylabel("Num", fontsize=20)

plt.hist(num, color="g", edgecolor="k", linewidth=2, linestyle="solid", cumulative=-1)
plt.show()
```



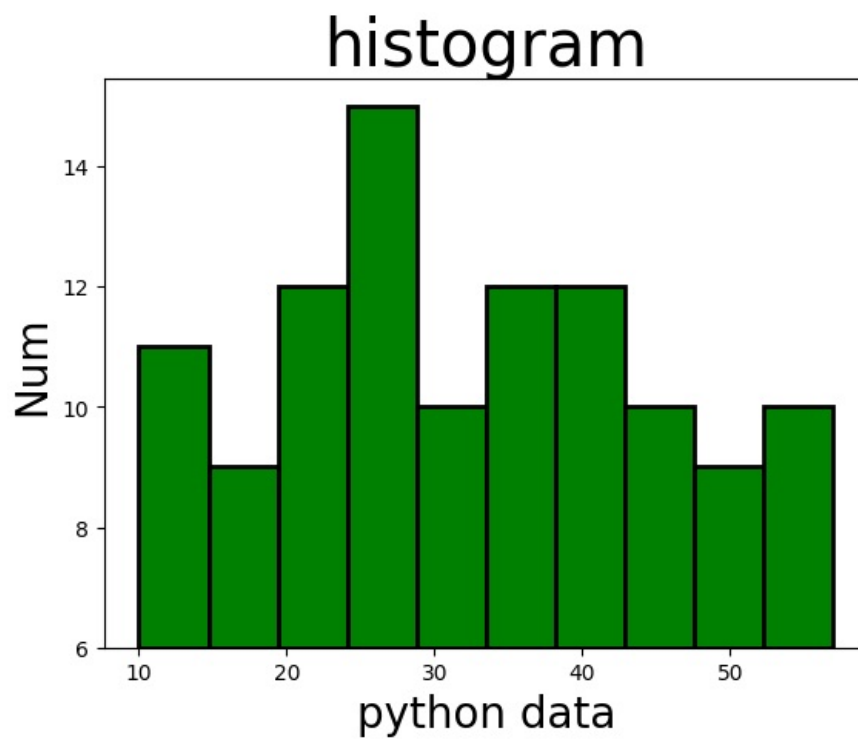
bottom parameter

```
In [42]: num=[38, 47, 41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]
```

```
plt.title("histogram", fontsize=30)
plt.xlabel("python data", fontsize=20)

plt.ylabel("Num", fontsize=20)

plt.hist(num, color="g", edgecolor="k", linewidth=2, linestyle="solid", bottom=6)
plt.show()
```

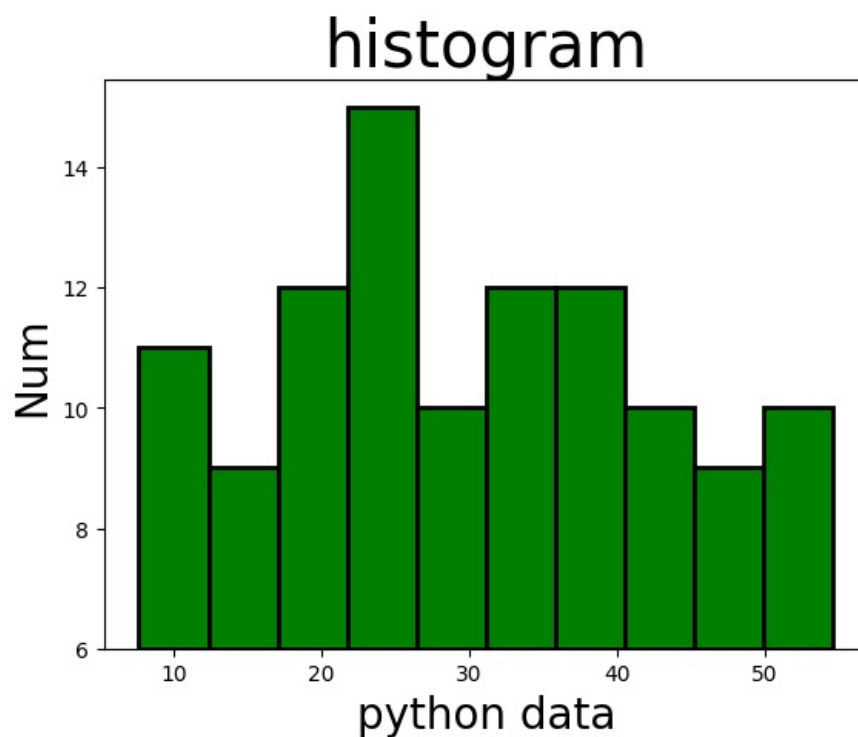
Left align

```
In [43]: num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6,align="left")
plt.show()
```



Hist-type

```
In [2]: import matplotlib.pyplot as plt
```

```

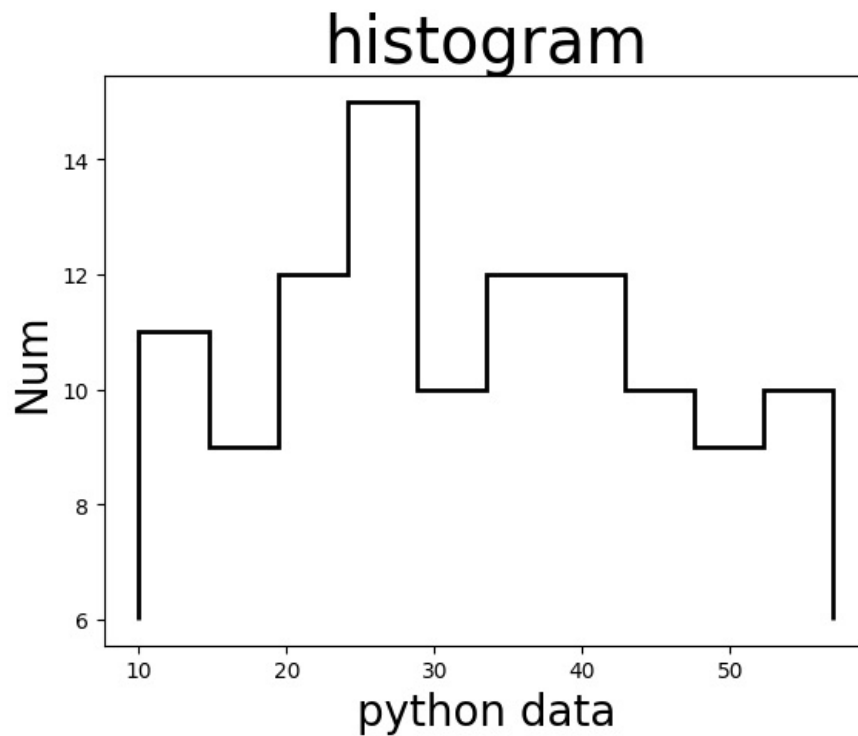
num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6,histtype="step")
#histype=step,stepfilled,barstackded,bar.
plt.show()

```



Orientation

```

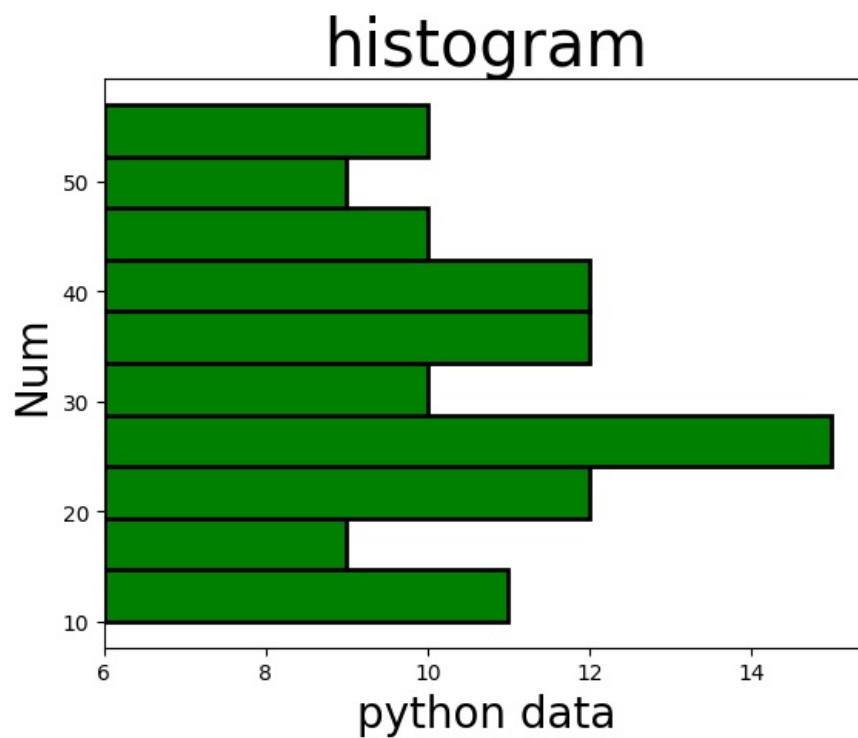
In [3]: import matplotlib.pyplot as plt
num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6,orientation="horizontal")
#
plt.show()

```



width between the hist

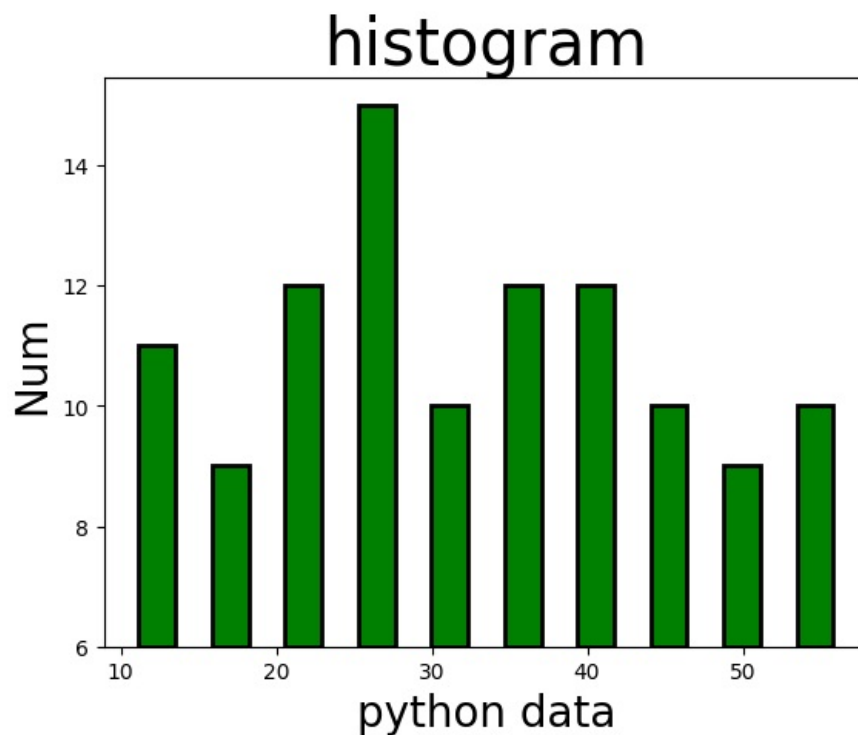
```
In [6]: import matplotlib.pyplot as plt
num=[38, 47, 41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram", fontsize=30)
plt.xlabel("python data", fontsize=20)

plt.ylabel("Num", fontsize=20)

plt.hist(num, color="g", edgecolor="k", linewidth=2, linestyle="solid", bottom=6, rwidth=0.5)

plt.show()
```



convert into log

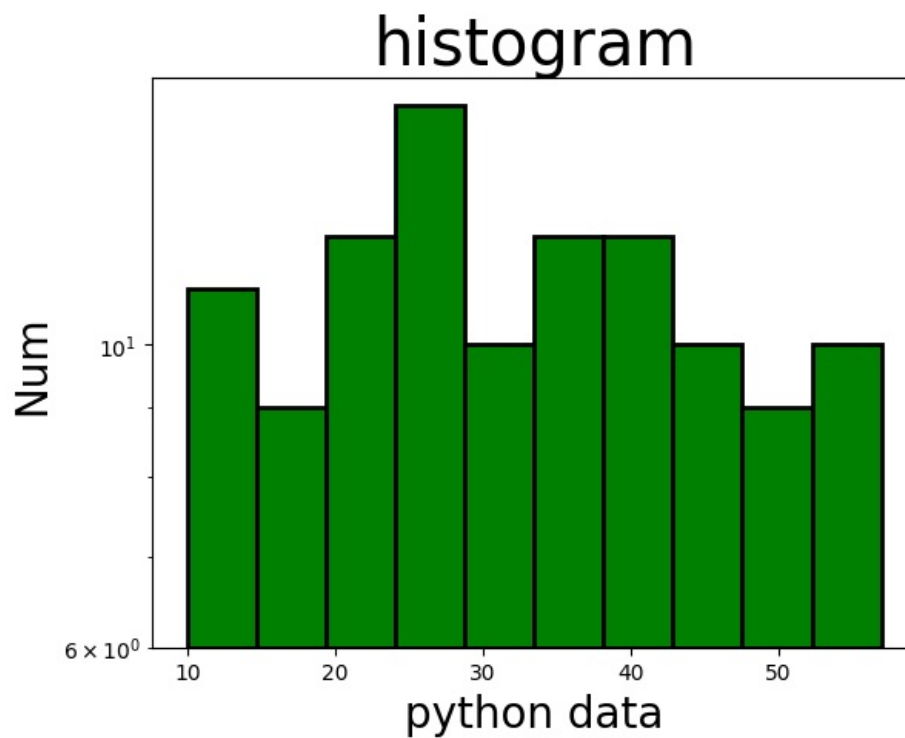
```
In [9]: import matplotlib.pyplot as plt
num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6,log=True)

plt.show()
```



plt.axvline()

```
In [16]: import matplotlib.pyplot as plt
num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

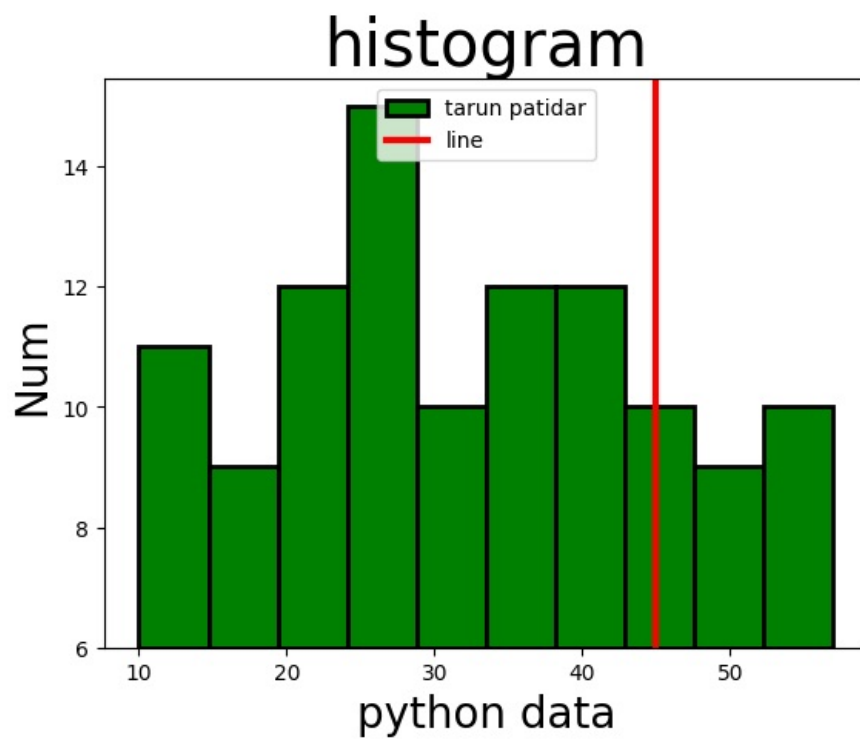
plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6,label="tarun patidar")

plt.axvline(45,color="r", linewidth=3,label="line")
plt.legend(loc="upper center")

plt.show()
```



plt.grid()

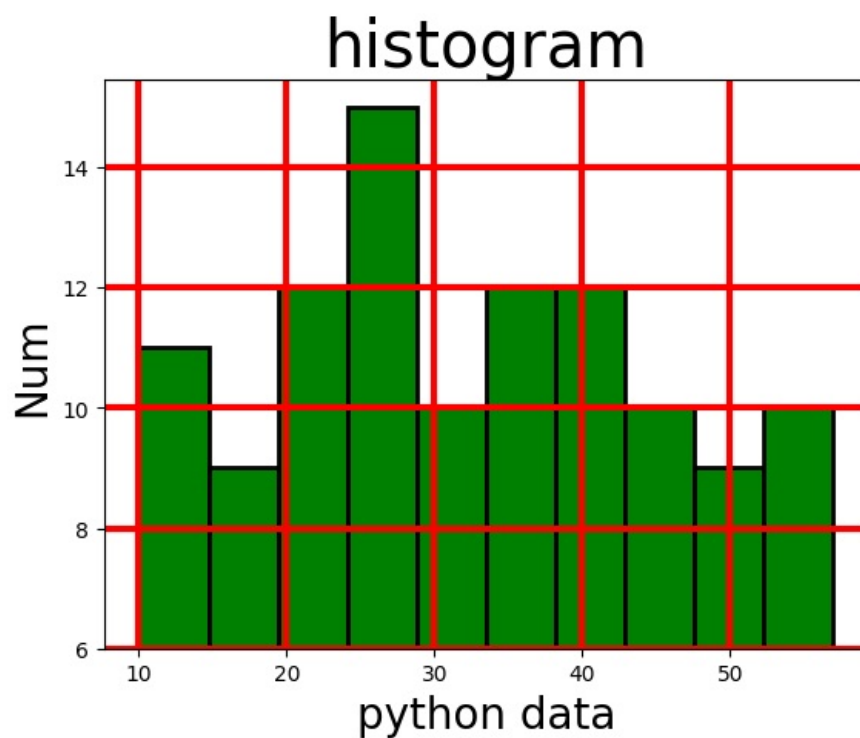
```
In [18]: import matplotlib.pyplot as plt
num=[38, 47,41, 26, 25, 48, 54, 28, 14, 36, 42, 40, 32, 22, 49, 39, 17, 27, 10, 26, 57, 34, 10, 41,
29, 22, 26, 50, 36, 19, 37, 55, 10, 37, 45, 53, 42, 19, 20, 20, 11, 29, 27, 45, 32, 47, 23, 23,
25, 28]

plt.title("histogram",fontsize=30)
plt.xlabel("python data",fontsize=20)

plt.ylabel("Num",fontsize=20)

plt.hist(num,color="g",edgecolor="k",linewidth=2,linestyle="solid",bottom=6)
plt.grid(linewidth=3, color="r", linestyle="solid")

plt.show()
```



pie chart

```
In [25]: import matplotlib.pyplot as plt

x=[10,20,30,40]

plt.pie(x)
plt.show()
```

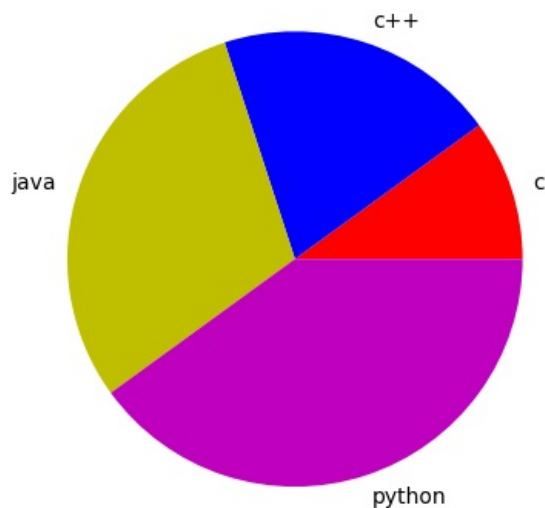


pie chart with labels & colors

```
In [31]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]
c=["r","b","y","m"]

plt.pie(x,labels=y,colors=c)
plt.show()
```



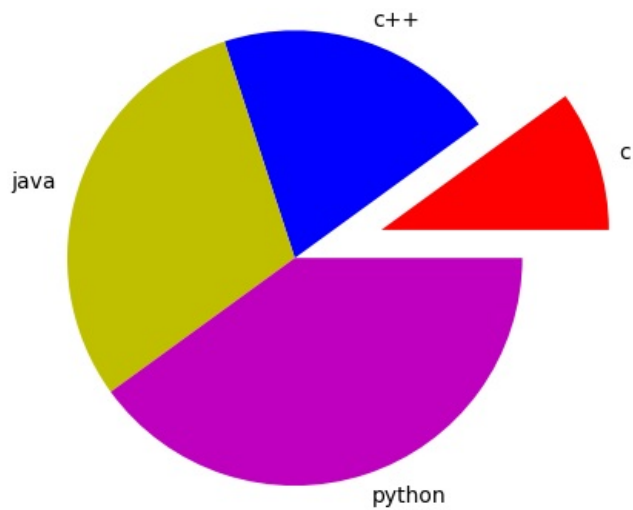
explode=explode

```
In [36]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]
c=["r","b","y","m"]

explode=[0.4,0,0,0]
```

```
plt.pie(x,labels=y,explode=explode,colors=c)
plt.show()
```



autopct="%0.1f%%"

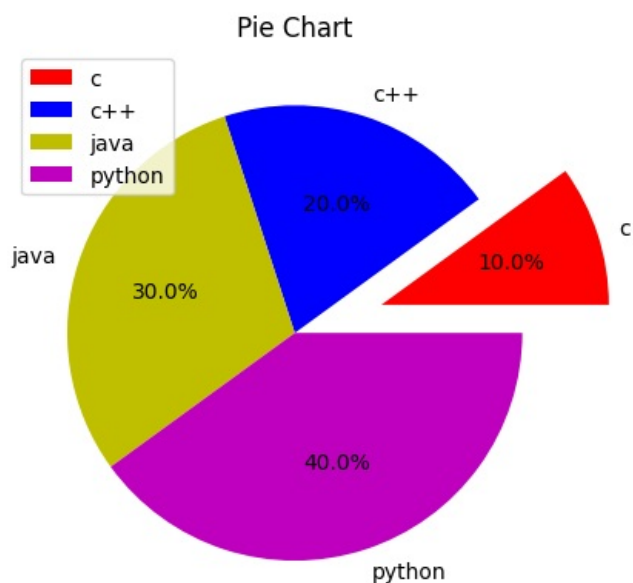
```
In [80]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]
c=["r","b","y","m"]

explode=[0.4,0,0,0]

plt.pie(x,labels=y,explode=explode,colors=c,autopct="%0.1f%%")
# 0.1f =>means decimal ke baad ek zero(0)

plt.title("Pie Chart")
plt.legend(loc="upper left")
plt.show()
```



shadow=True,radius=2

```
In [38]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]
```

```

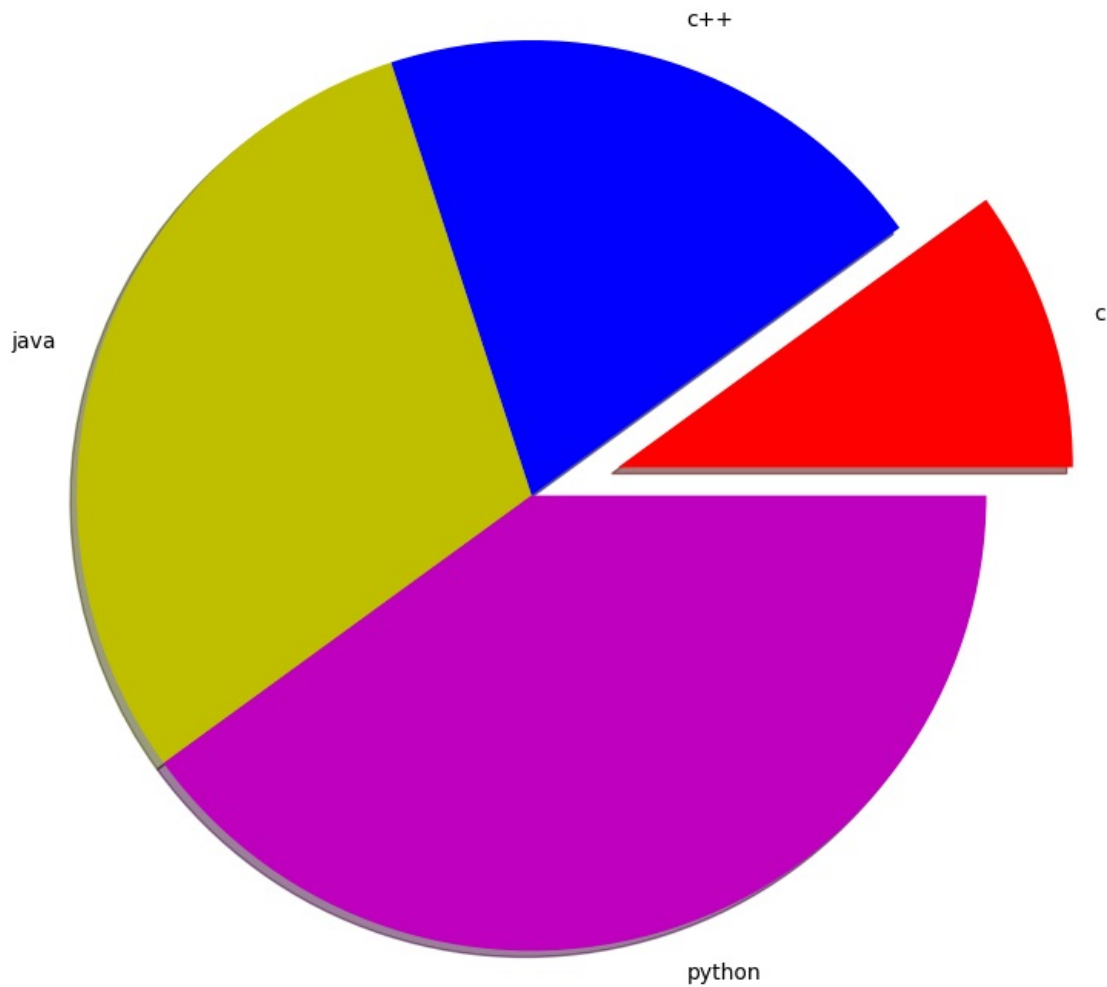
c=["r","b","y","m"]

explode=[0.4,0,0,0]

plt.pie(x,labels=y,explode=explode,colors=c,shadow=True,radius=2)

plt.show()

```



labeldistance=0.5

```

In [42]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]

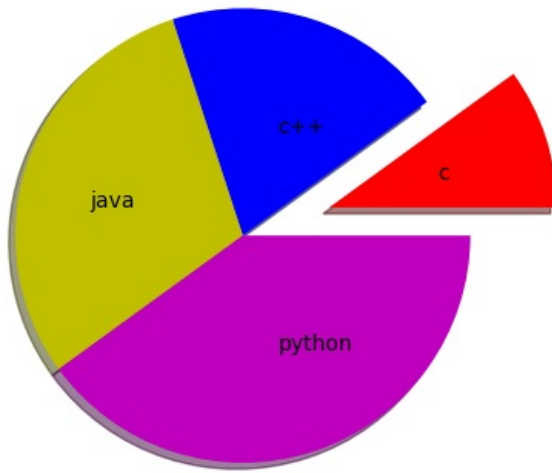
c=["r","b","y","m"]

explode=[0.4,0,0,0]

plt.pie(x,labels=y,explode=explode,colors=c,shadow=True,labeldistance=0.5)

plt.show()

```

startangle=90 => Start-angle

```
In [43]: import matplotlib.pyplot as plt

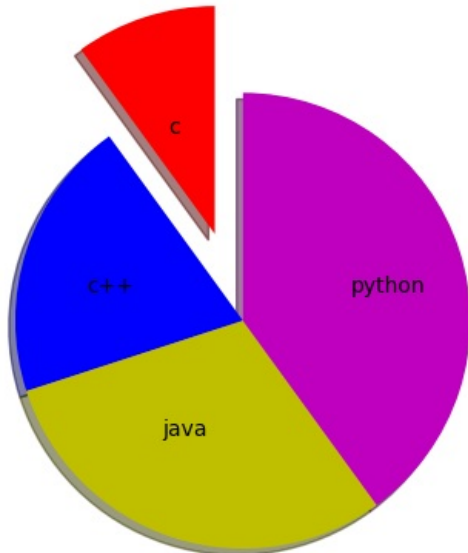
x=[10,20,30,40]
y=["c", "c++", "java", "python"]

c=["r", "b", "y", "m"]

explode=[0.4,0,0,0]

plt.pie(x, labels=y, explode=explode, colors=c, shadow=True, labeldistance=0.5, startangle=90)

plt.show()
```



text properties => textprops={key:value}

```
In [83]: import matplotlib.pyplot as plt

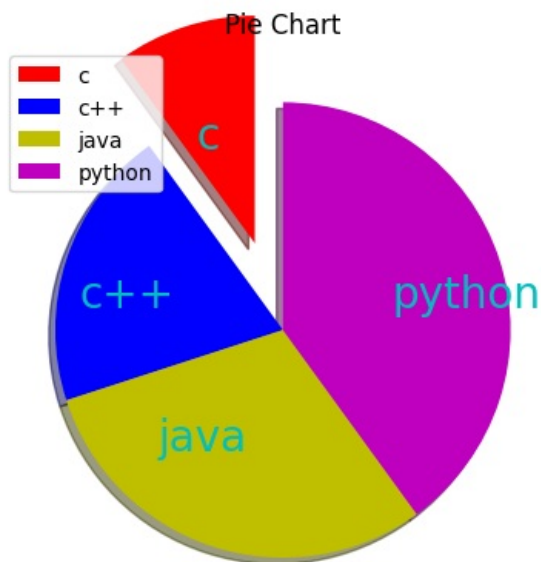
x=[10,20,30,40]
y=["c", "c++", "java", "python"]

c=["r", "b", "y", "m"]

explode=[0.4,0,0,0]

plt.pie(x, labels=y, explode=explode, colors=c, shadow=True, labeldistance=0.5, startangle=90,
        textprops={"fontsize":20, "color":"c"})
```

```
plt.title("Pie Chart")
plt.legend(loc="upper left")
# text properties => textprops={key:value}
plt.show()
```



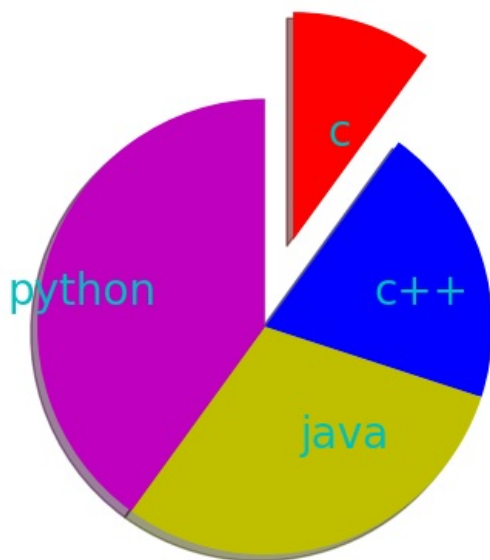
counterclock=False

```
In [82]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c","c++","java","python"]
c=["r","b","y","m"]
explode=[0.4,0,0,0]

plt.pie(x,labels=y,explode=explode,colors=c,shadow=True,labeldistance=0.5,startangle=90,
        textprops={"fontsize":20,"color":"c"},counterclock=False)

# text properties => textprops={key:value}
plt.show()
```



edge width properties => wedgeprops = {key:value}

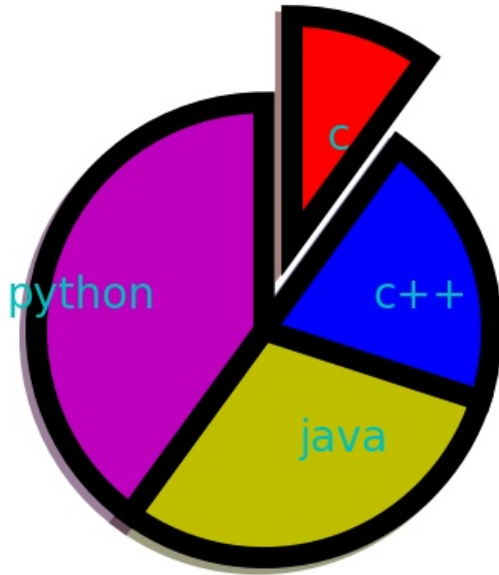
```
In [69]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c", "c++", "java", "python"]
c=["r", "b", "y", "m"]

explode=[0.4,0,0,0]

plt.pie(x, labels=y, explode=explode, colors=c, shadow=True, labeldistance=0.5, startangle=90,
        textprops={"fontsize":20, "color": "c"}, counterclock=False, wedgeprops={"linewidth":10, "edgecolor": "k"})

#   edge width properties => wedgeprops = {key:value}
plt.show()
```



wedgeprops=
{"linewidth":10,"width":2,"edgecolor":"k"}

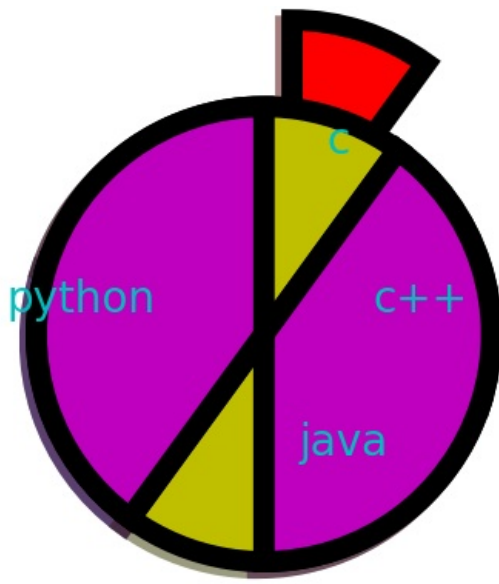
```
In [70]: import matplotlib.pyplot as plt

x=[10,20,30,40]
y=["c", "c++", "java", "python"]
c=["r", "b", "y", "m"]

explode=[0.4,0,0,0]

plt.pie(x, labels=y, explode=explode, colors=c, shadow=True, labeldistance=0.5, startangle=90,
        textprops={"fontsize":20, "color": "c"}, counterclock=False, wedgeprops={"linewidth":10, "width":2, "edgecolor": "k"},
        wedgeprops={"linewidth":10, "width":2, "edgecolor": "k"})

#   edge width properties => wedgeprops = {key:value}
plt.show()
```



center=(3,6))

```
In [72]: import matplotlib.pyplot as plt

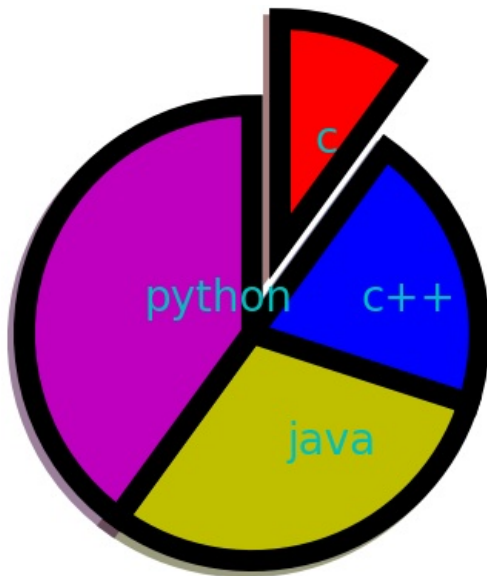
x=[10,20,30,40]
y=["c", "c++", "java", "python"]

c=["r", "b", "y", "m"]

explode=[0.4,0,0,0]

plt.pie(x, labels=y, explode=explode, colors=c, shadow=True, labeldistance=0.5, startangle=90,
        textprops={"fontsize":20, "color": "c"}, counterclock=False, wedgeprops={"linewidth":10, "edgecolor": "k"},
        center=(3,6))

#   edge width properties => wedgeprops = {key:value}
plt.show()
```



rotatelabels=True

```
In [76]: import matplotlib.pyplot as plt

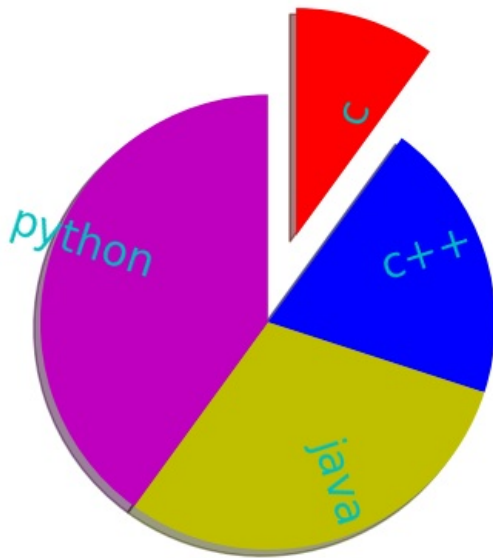
x=[10,20,30,40]
y=["c", "c++", "java", "python"]

c=["r", "b", "y", "m"]
```

```
explode=[0.4,0,0,0]
```

```
plt.pie(x,labels=y,explode=explode,colors=c,shadow=True,labeldistance=0.5,startangle=90,  
        textprops={"fontsize":20,"color":"c"},counterclock=False,rotatelabels=True)
```

```
# text properties => textprops={key:value}  
plt.show()
```



plt.legend(loc=2)

```
In [90]: import matplotlib.pyplot as plt
```

```
x=[10,20,30,40]
```

```
y=["c","c++","java","python"]
```

```
c=["r","b","y","m"]
```

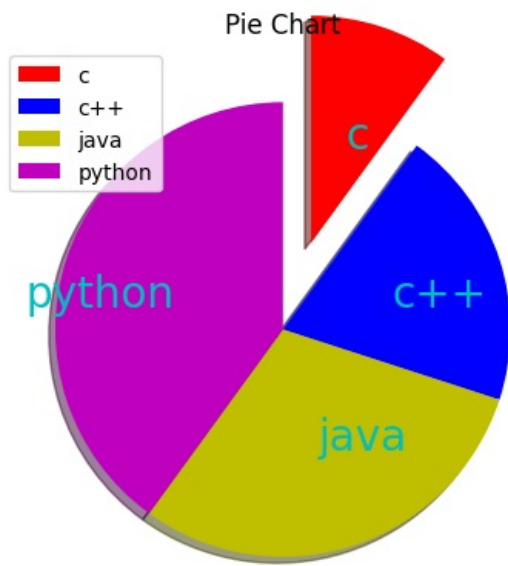
```
explode=[0.4,0,0,0]
```

```
plt.pie(x,labels=y,explode=explode,colors=c,shadow=True,labeldistance=0.5,startangle=90,  
        textprops={"fontsize":20,"color":"c"},counterclock=False,rotatelabels=False,)
```

```
plt.title("Pie Chart")
```

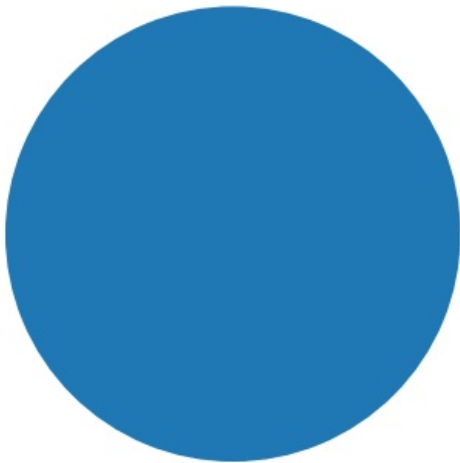
```
plt.legend(loc=2)
```

```
# text properties => textprops={key:value}  
plt.show()
```



dot pie chart

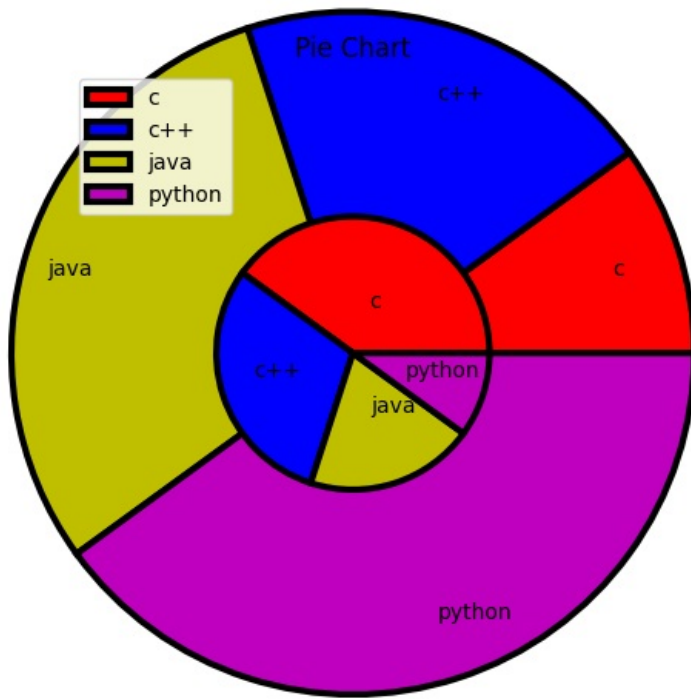
```
In [91]: plt.pie([1])  
plt.show()
```



Multiple Pie Chart

```
In [113]: import matplotlib.pyplot as plt  
  
x=[10,20,30,40]  
x2=[40,30,20,10]  
y=["C", "C++", "java", "python"]  
c=["r", "b", "y", "m"]  
  
explode=[0.4,0,0,0]  
  
plt.pie(x, labels=y, colors=c, wedgeprops={"linewidth":3, "edgecolor":"k"},  
        labeldistance=0.8, radius=1.5) # outer pie chart  
  
plt.legend(loc="upper left")  
  
plt.pie(x2, labels=y, colors=c, wedgeprops={"linewidth":3, "edgecolor":"k"},  
        labeldistance=0.4, radius=0.6) # inner pie chart  
  
plt.title("Pie Chart")
```

```
# text properties => textprops={key:value}
plt.show()
```



Multiple Pie Chart

```
In [117... import matplotlib.pyplot as plt

x=[10,20,30,40]
x2=[40,30,20,10]
y=["c","c++","java","python"]
c=["r","b","y","m"]

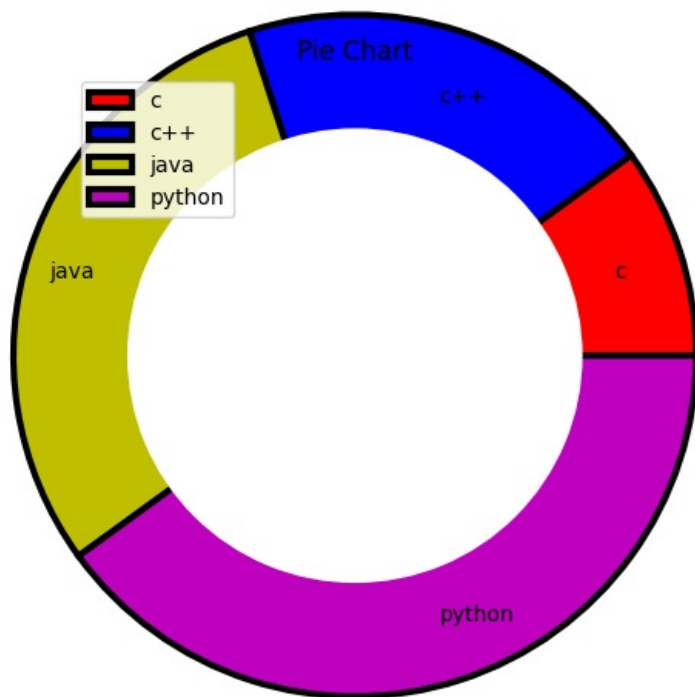
explode=[0.4,0,0,0]

plt.pie(x,labels=y,colors=c,wedgeprops={"linewidth":3,"edgecolor":"k"},
        labeldistance=0.8,radius=1.5) # outer pie chart

plt.legend(loc="upper left")

plt.pie([1],colors="w",radius=1) # inner pie chart
plt.title("Pie Chart")

# text properties => textprops={key:value}
plt.show()
```



```
cir=plt.Circle(xy=(0,0),radius=1,facecolor="w")
```

```
plt.gca().add_artist(cir)
```

```
In [119]: import matplotlib.pyplot as plt

x=[10,20,30,40]
x2=[40,30,20,10]
y=["c","c++","java","python"]

c=["r","b","y","m"]

explode=[0.4,0,0,0]

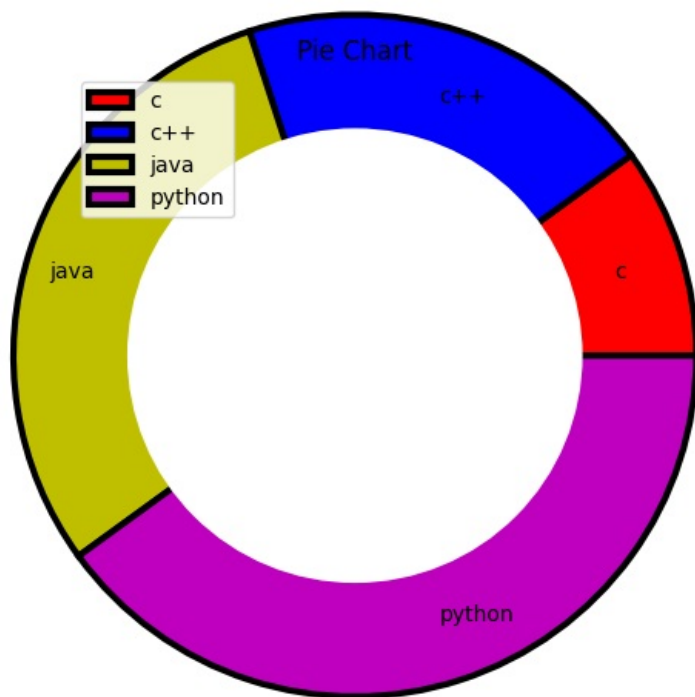
plt.pie(x,labels=y,colors=c,wedgeprops={"linewidth":3,"edgecolor":"k"},
        labeldistance=0.8,radius=1.5) # outer pie chart

plt.legend(loc="upper left")

cir=plt.Circle(xy=(0,0),radius=1,facecolor="w")
plt.gca().add_artist(cir)

plt.title("Pie Chart")

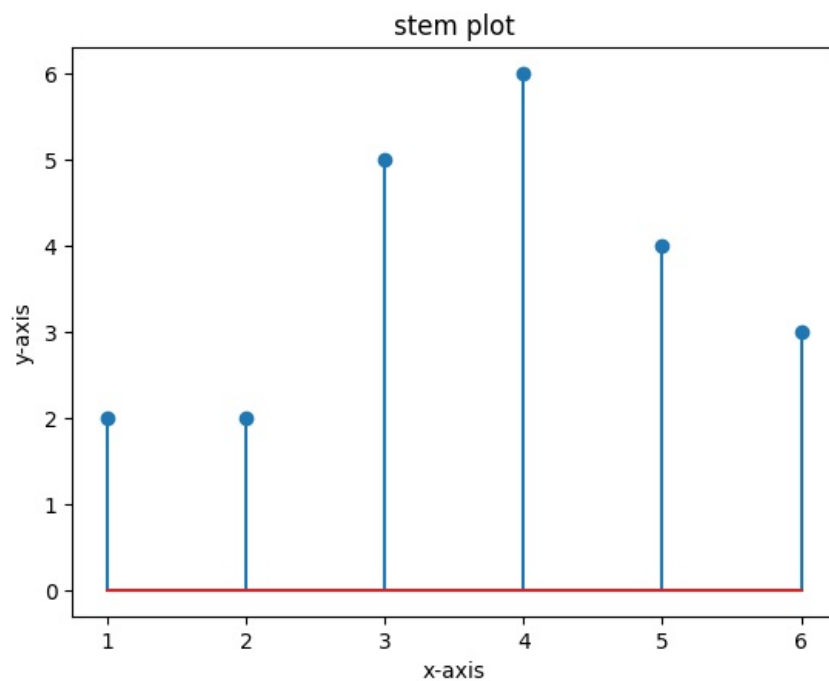
# text properties => textprops={key:value}
plt.show()
```

stem() plot

```
In [122]: import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

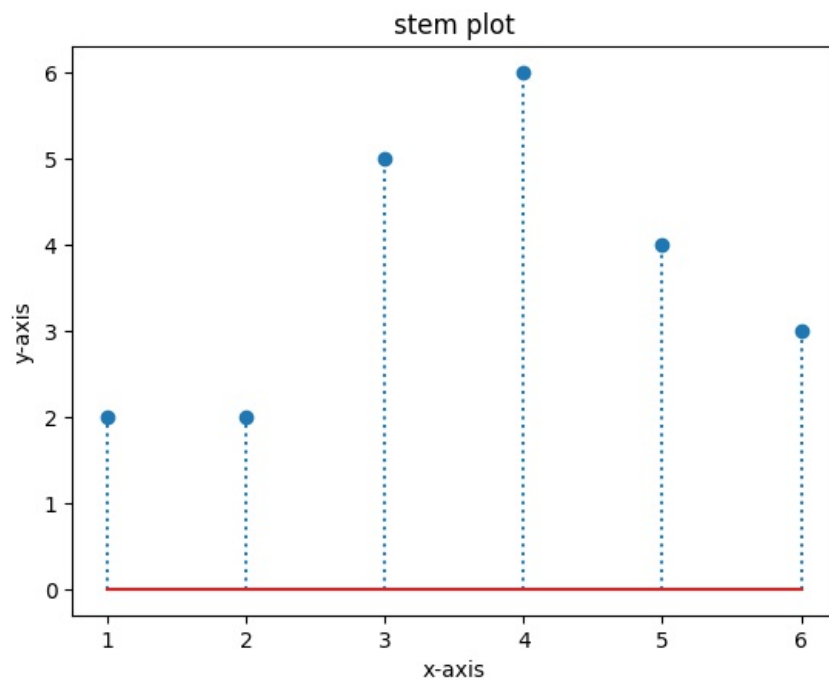
plt.stem(x,y)
plt.show()
```



linefmt=":"

```
In [123]: import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

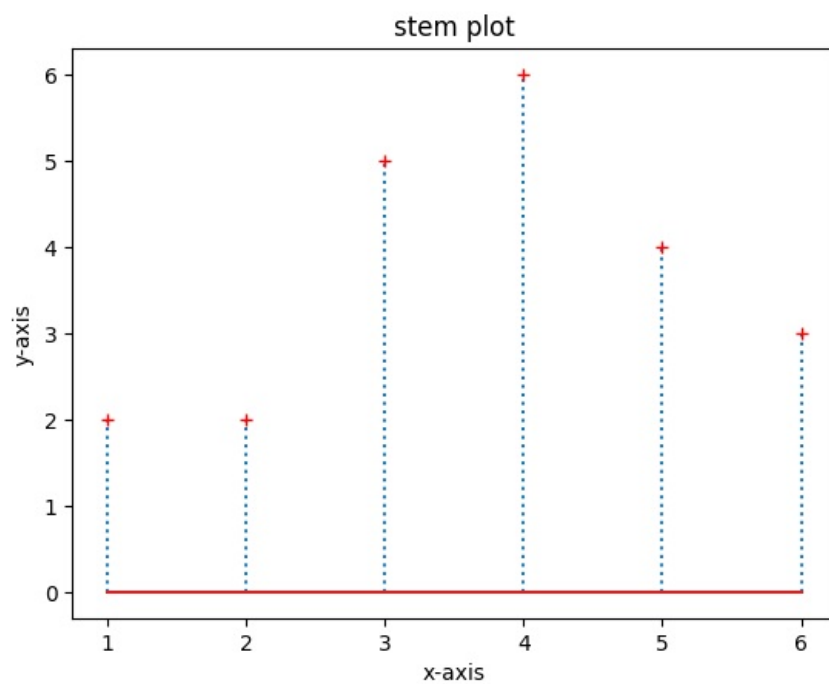
plt.stem(x,y,linefmt=":")
plt.show()
```



markerfmt="r+"

```
In [124... import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

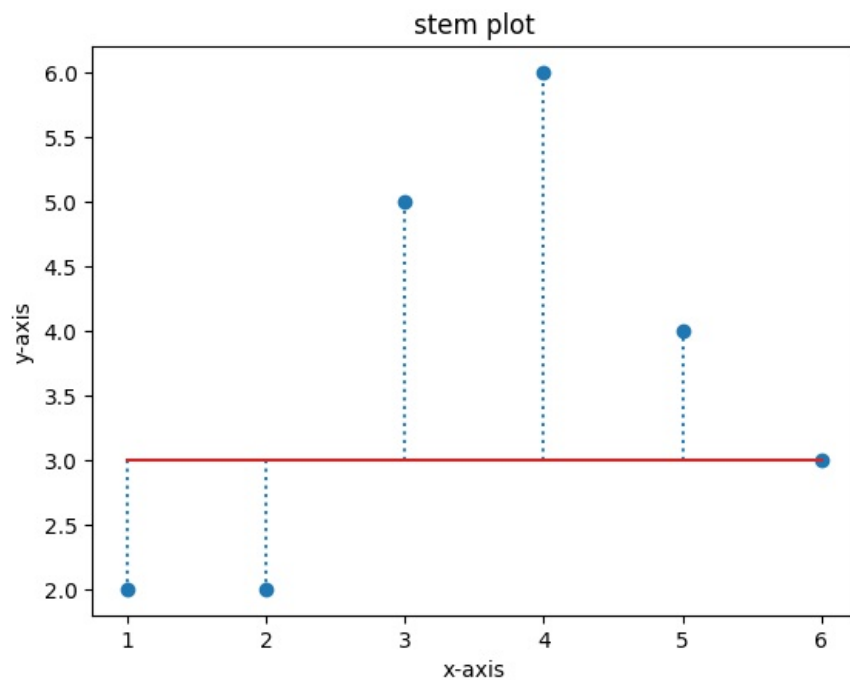
plt.stem(x,y,linefmt=":",markerfmt="r+")
plt.show()
```



bottom=3

```
In [127... import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

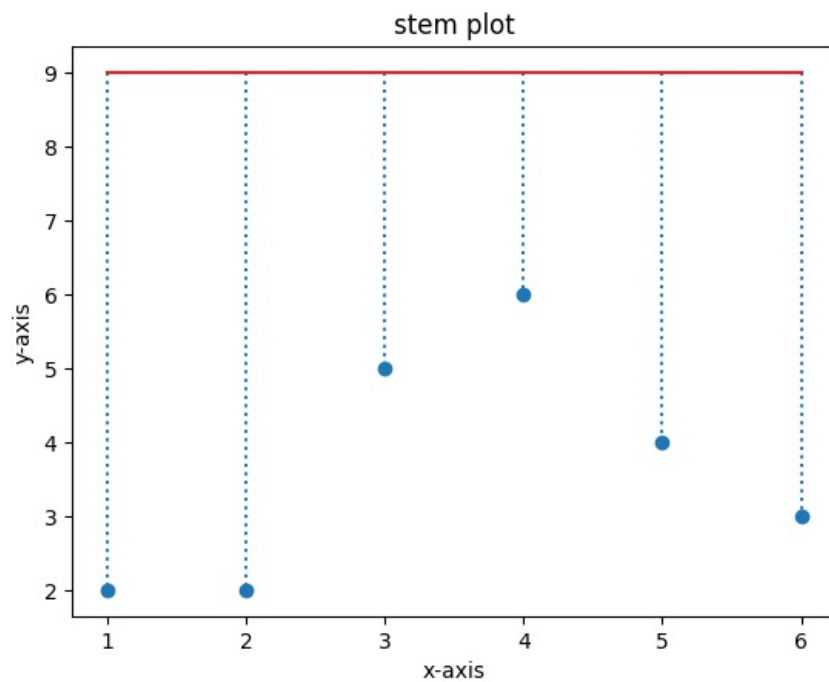
plt.stem(x,y,linefmt=":",bottom=3)
plt.show()
```



bottom=9

```
In [130.. import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

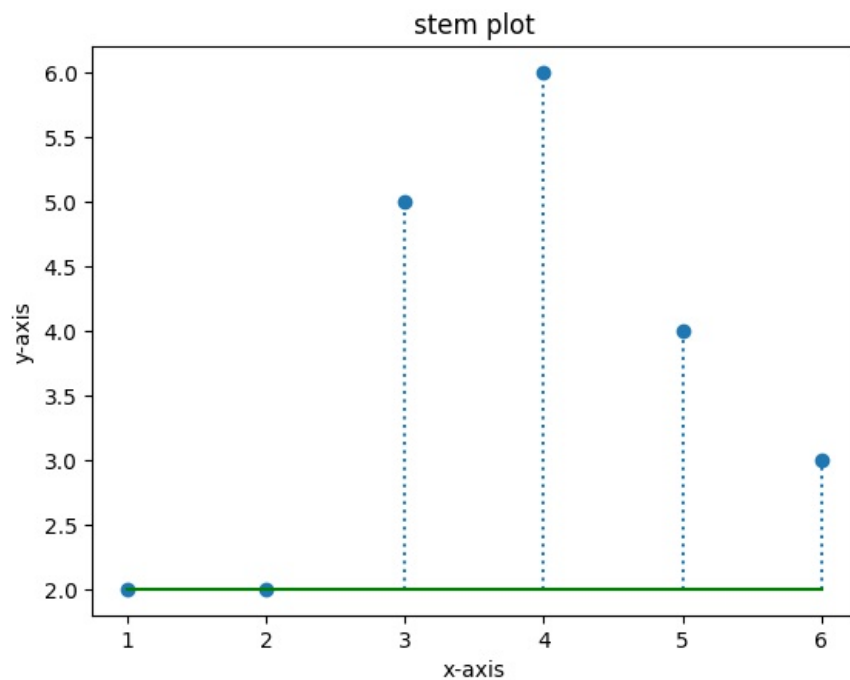
plt.stem(x,y,linefmt=":",bottom=9)
plt.show()
```



basefmt="g"

```
In [133.. import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

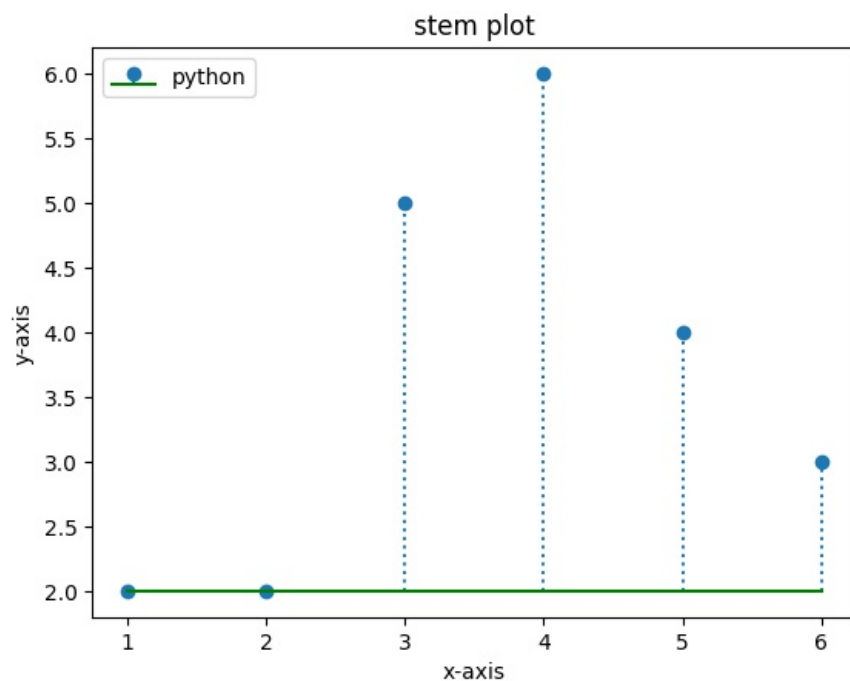
plt.stem(x,y,linefmt=":",bottom=2,basefmt="g")
plt.show()
```



plt.legend(loc="upper left")

```
In [136]: import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

plt.stem(x,y, linefmt=":", bottom=2, basefmt="g", label="python")
plt.legend(loc="upper left")
plt.show()
```

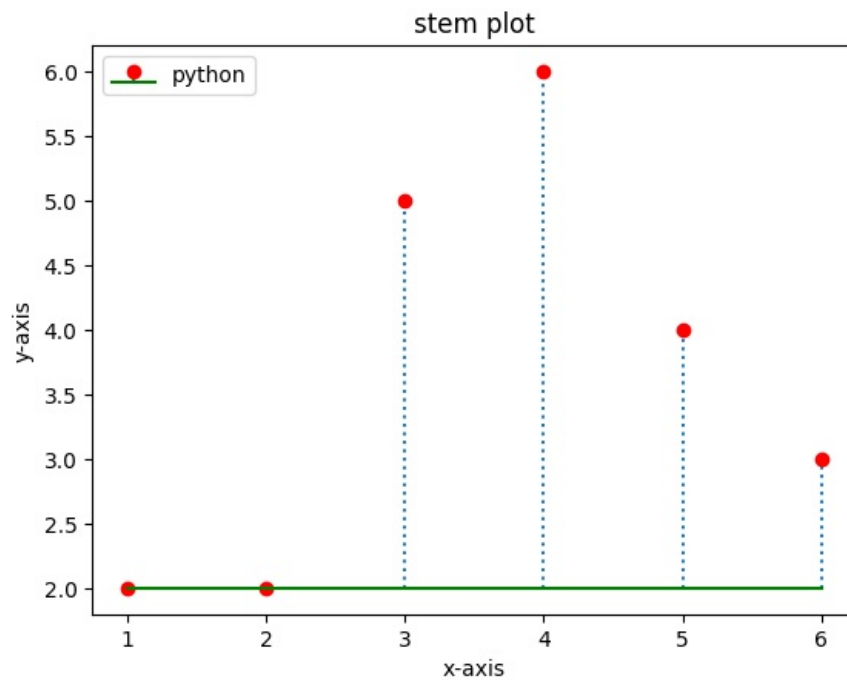


plt.legend(loc="upper left")

```
In [139]: import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

plt.stem(x,y, linefmt=":", markerfmt="ro", bottom=2, basefmt="g", label="python")
plt.legend(loc="upper left")
```

```
plt.show()
```

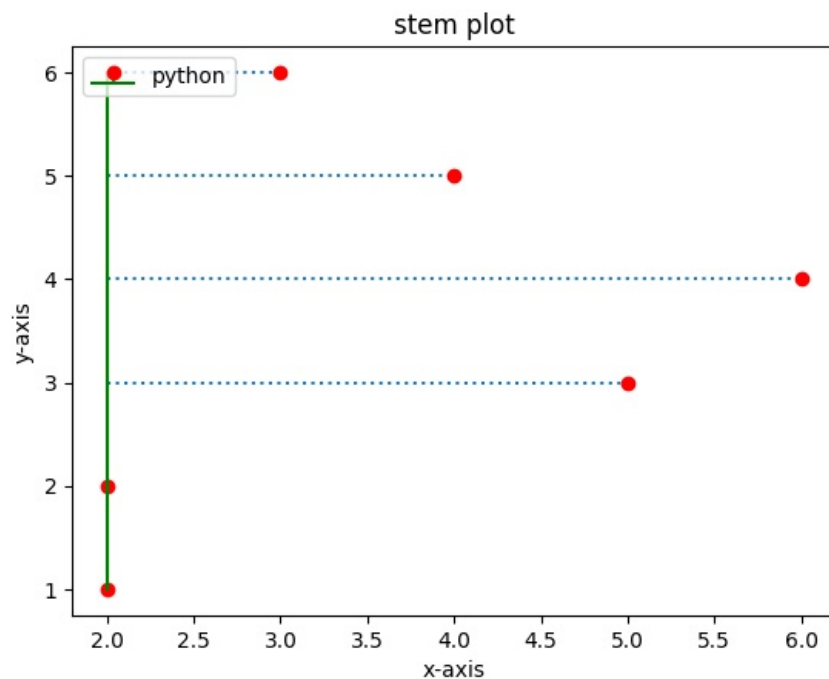


orientation="horizontal"

```
In [146.. import matplotlib.pyplot as plt
x=[1,2,3,4,5,6]
y=[2,2,5,6,4,3]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("stem plot")

plt.stem(x,y,linefmt=":",markerfmt="ro",bottom=2,basefmt="g",
        label="python",orientation="horizontal")

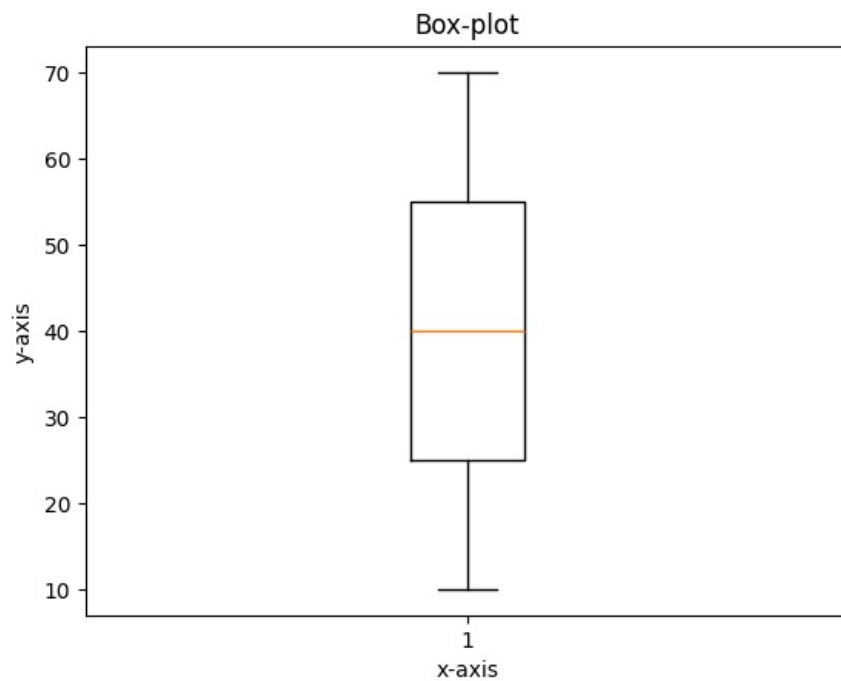
plt.legend(loc="upper left")
plt.show()
```



Boxplot()

```
In [149.. import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")
```

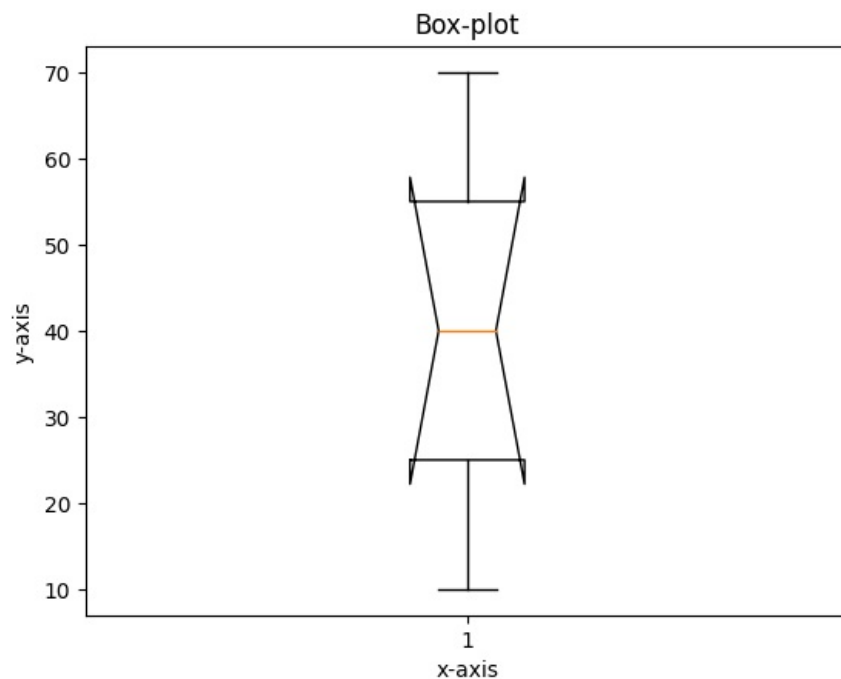
```
plt.boxplot(x)
plt.show()
```



notch=True

```
In [153... import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

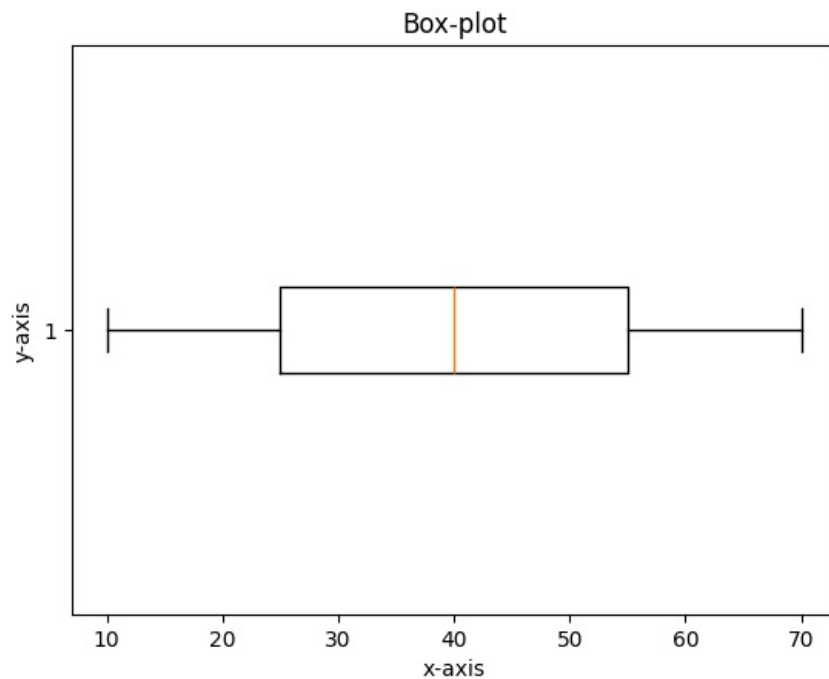
plt.boxplot(x,notch=True)
plt.show()
```



Horizontal Box-plot

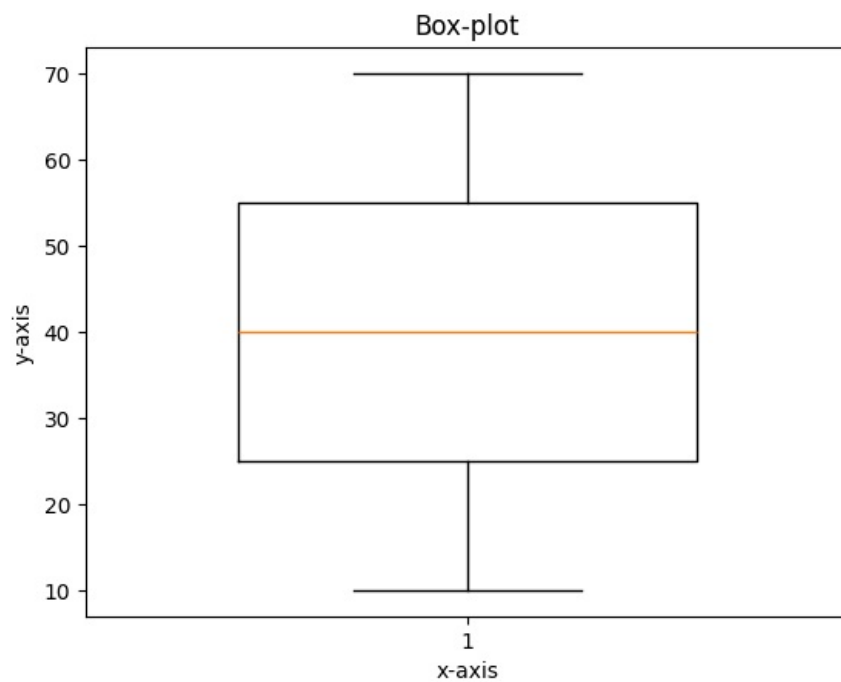
```
In [154... import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")
```

```
plt.boxplot(x,vert=False)  
plt.show()
```



widths=0.6

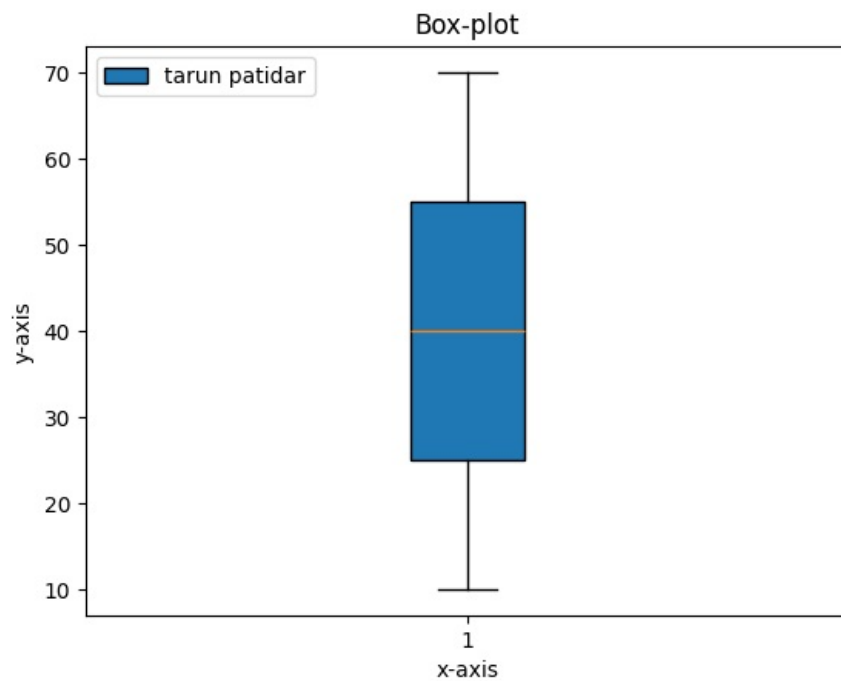
```
In [156.. import matplotlib.pyplot as plt  
x=[10,20,30,40,50,60,70]  
plt.xlabel("x-axis")  
plt.ylabel("y-axis")  
plt.title("Box-plot")  
  
plt.boxplot(x,widths=0.6)  
plt.show()
```



patch_artist=True

```
In [159.. import matplotlib.pyplot as plt  
x=[10,20,30,40,50,60,70]  
plt.xlabel("x-axis")  
plt.ylabel("y-axis")  
plt.title("Box-plot")
```

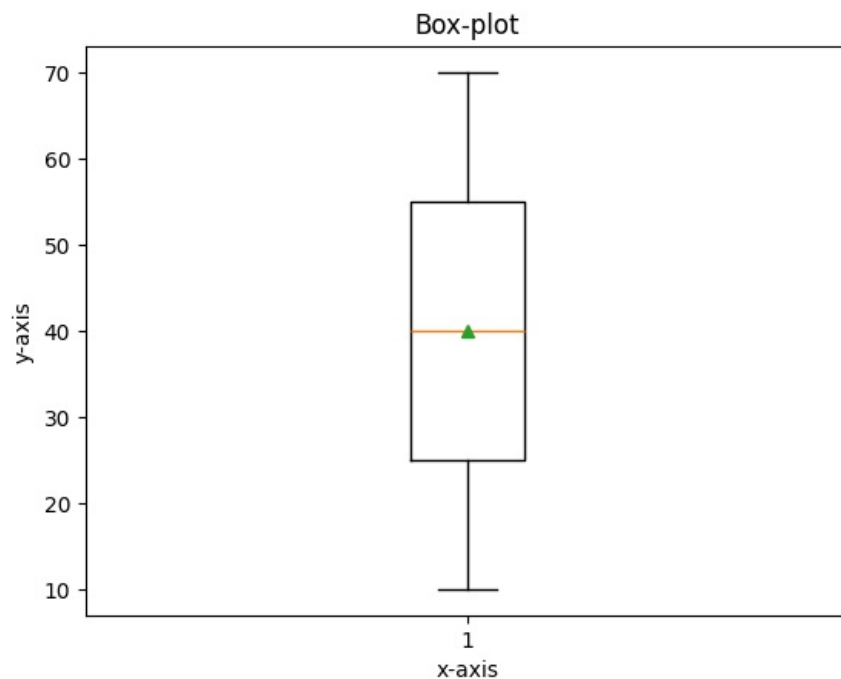
```
plt.boxplot(x,label=["tarun patidar"],patch_artist=True)
plt.legend(loc="upper left")
plt.show()
```



showmeans=True

```
In [161]: import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

plt.boxplot(x,showmeans=True)
plt.show()
```

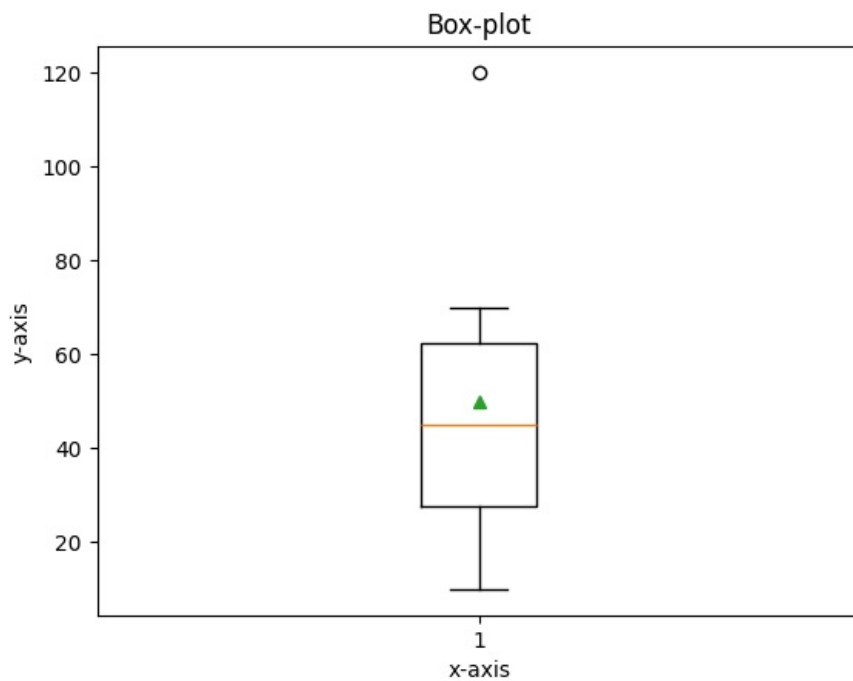


Outlier

```
In [162]: import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")
```



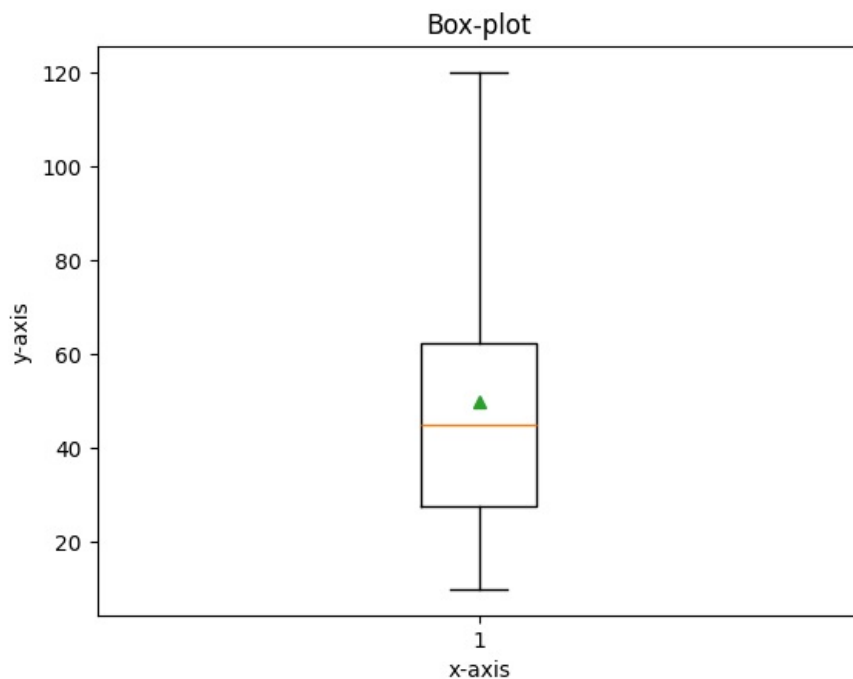
```
plt.boxplot(x, showmeans=True)
plt.show()
```



whis => joint to the outlier

```
In [163.. import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

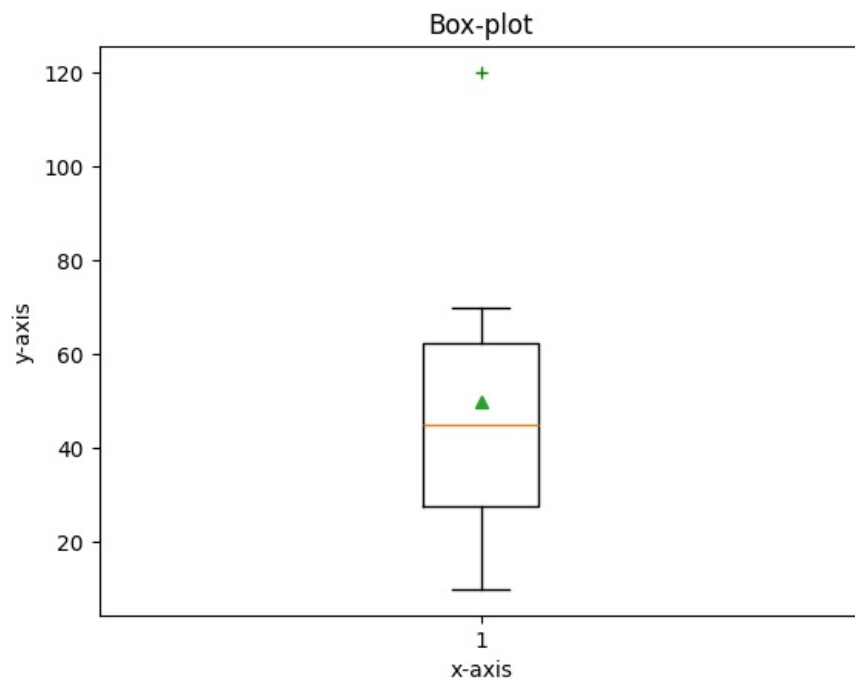
plt.boxplot(x, showmeans=True, whis=3.5)
plt.show()
```



sym="g+" => change the outlier color

```
In [164.. import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")
```

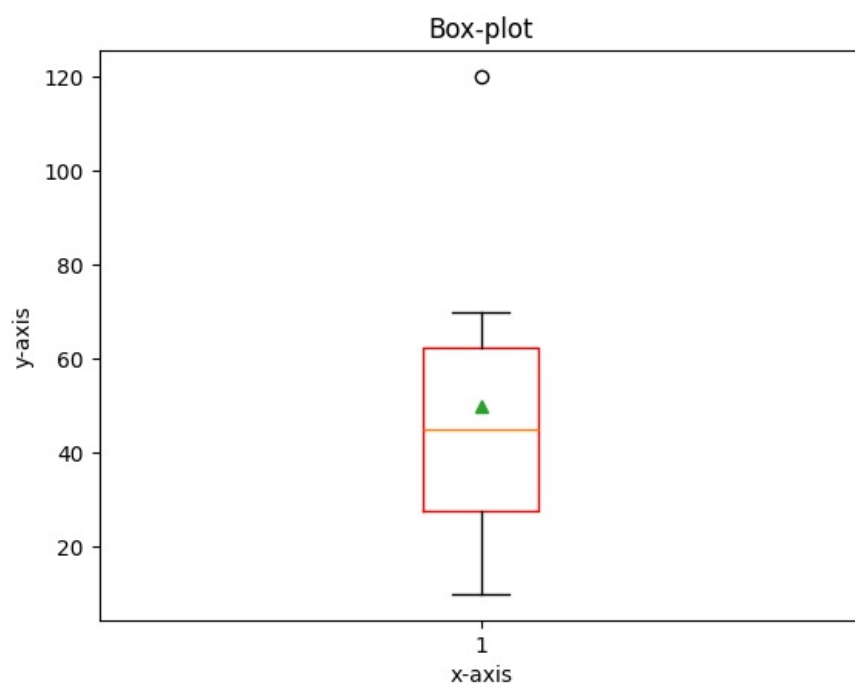
```
plt.boxplot(x, showmeans=True, sym="g+")
plt.show()
```



boxprops=dict(color="r",linewidth=4) => box-properties

```
In [165... import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

plt.boxplot(x, showmeans=True, boxprops=dict(color="r"))
plt.show()
```

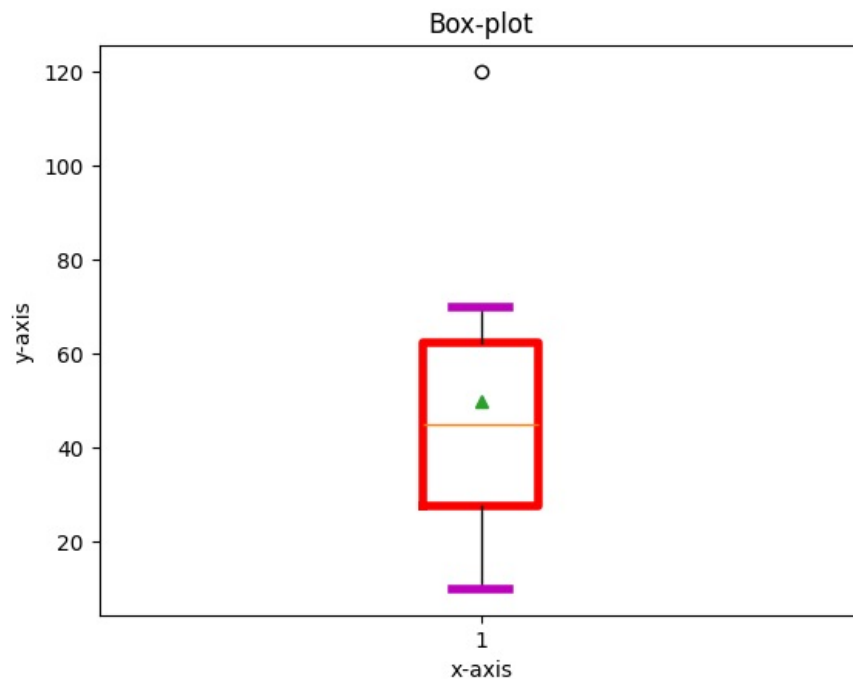


capprops=dict(color="m",linewidth=4)

```
In [171... import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
```

```
plt.ylabel("y-axis")
plt.title("Box-plot")

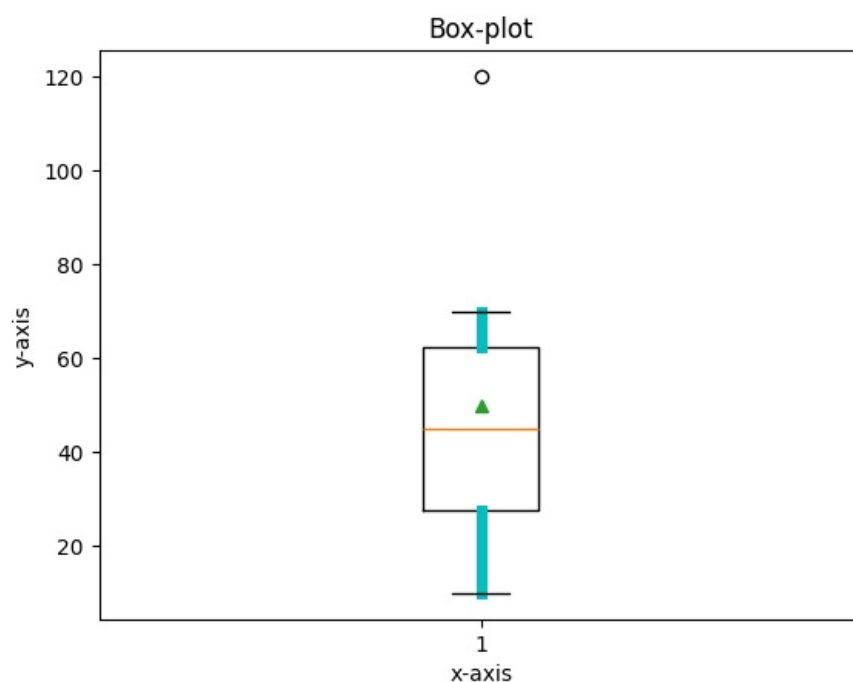
plt.boxplot(x, showmeans=True, boxprops=dict(color="r", linewidth=4), capprops=dict(color="m", linewidth=4))
#cap - properties
# box- properties
plt.show()
```



whiskerprops=dict(color="c",linewidth=5) ==> whisker properties

```
In [174]: import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

plt.boxplot(x, showmeans=True, whiskerprops=dict(color="c", linewidth=5))
plt.show()
```



flierprops=dict(marker='o', markerfacecolor='red',

**markedgedgecolor='blue', markersize=30,
linewidth=20)**

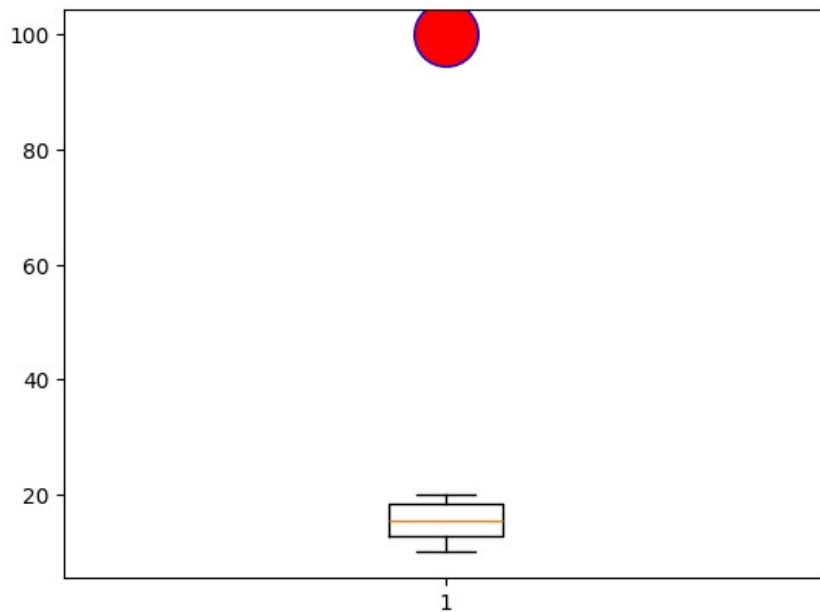
```
In [190... import matplotlib.pyplot as plt

x = [10, 12, 13, 15, 16, 18, 20, 100]

plt.boxplot(x,

            flierprops=dict(
                marker='o',
                markerfacecolor='red',
                markedgedgecolor='blue',
                markersize=30,
                linewidth=20
            )
        )

plt.show()
```



markersize=35, markedgedgecolor='blue',

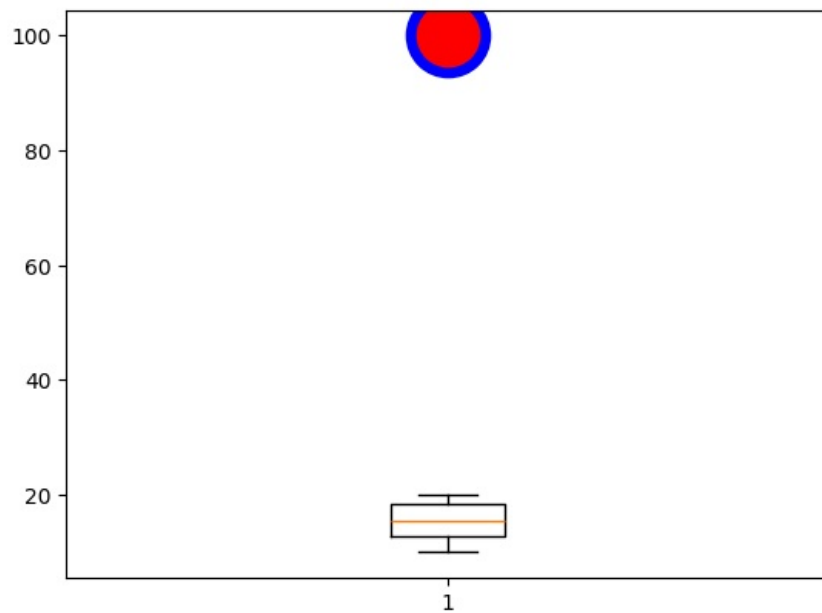
```
In [193... import matplotlib.pyplot as plt

x = [10, 12, 13, 15, 16, 18, 20, 100]

plt.boxplot( x,

            flierprops=dict(
                marker='o',
                markerfacecolor='red',
                markedgedgecolor='blue',
                markersize=35,
                markedgedgecolor='blue',
                markedgedgecolor='blue'
            )
        )

plt.show()
```



Multiple box-plot

```
In [199]: import matplotlib.pyplot as plt
x=[10,20,30,40,50,60,70,120]
x2=[10,20,40,50,30,60,90]
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Box-plot")

y=[x,x2]

plt.boxplot(y,tick_labels=["python","c++"],showmeans=True,sym="g+",boxprops=dict(color="r",
        linewidth=4),capprops=dict(color="m",linewidth=4),
        whiskerprops=dict(color="c",linewidth=5)
        )

plt.xticks(fontsize=16 ,color="r")

plt.show()
```

